

CHOZ

User Guide

Version 1.3.16 · Linux (ALSA / JACK / PipeWire)

About this document. This guide documents choz 1.3.16. It is written for the terminal application; choz's effects and its four artifacts (arpeggiator, step sequencer, metronome and arranger) are also published as a CLAP plugin for other hosts, and choz itself runs inside a DAW as a CLAP instrument — see section 2.5.

Screenshots were taken from choz running on Linux with the JACK/PipeWire backend. Device, bank and preset names in them belong to that installation and will differ on yours.

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1. Welcome to choz

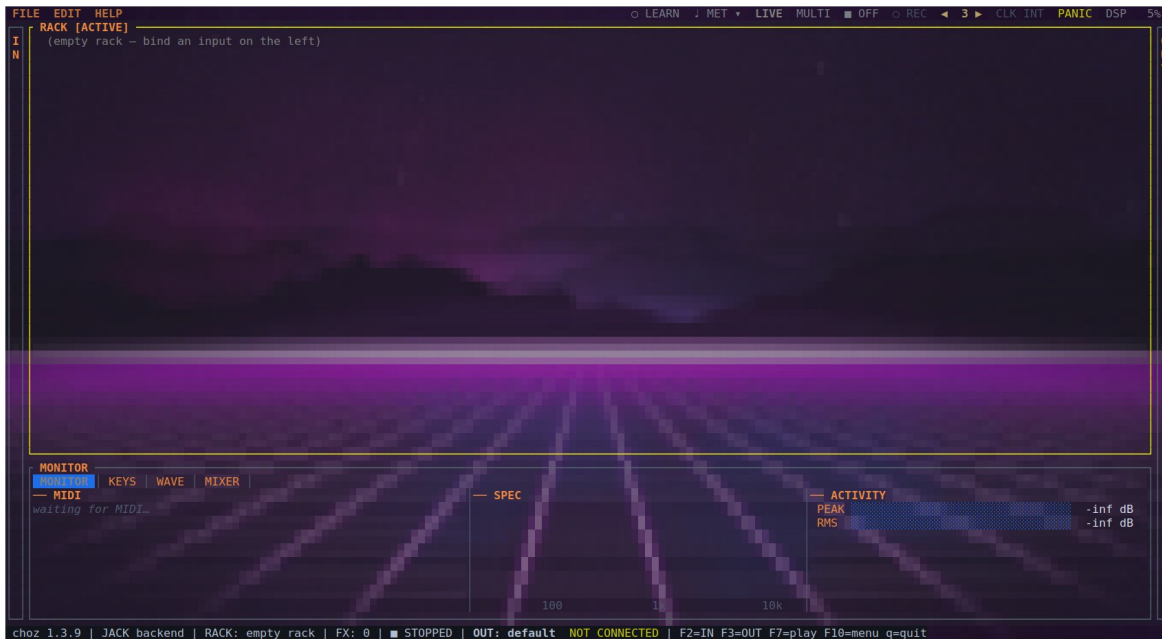


Figure 1.1 — choz just opened: the menu bar, an empty RACK, the IN and OUT drawers at the edges, and the MONITOR panel along the bottom.

1.1 What choz is

choz is an audio plugin host for the terminal. It gives you a rack of tabs; each tab takes notes or audio in, loads an instrument, chains effects onto it, and leaves either through the main mix or through a port of its own on the JACK/PipeWire graph.

- **Instruments:** SoundFont (SF2), SFZ, WAV, and CLAP, LV2, VST2, VST3, DSSI and LADSPA plugins, plus Pure Data patches.
- **Effects:** 56 of choz's own, plus every effect plugin installed on the machine, up to twelve per tab.
- **Note sources:** any MIDI port, choz's own ALSA and JACK ports, OSC, the computer keyboard, an arpeggiator and a step sequencer per tab, and audio-to-note tracking from a microphone.
- **Outputs:** the main stereo mix, four groups, and one direct out per tab for recording each tab on its own track in a DAW.

1.2 About this guide

Chapters 1 to 3 are the introduction: what choz needs, how to install it, and what is on the screen. Chapter 2 is a short walk-through that gets a sound out of it. Chapters 4 to 13 are the reference, one chapter per area of the program. Chapters 14 and 15 are lookup tables — shortcuts, environment variables, files. Chapter 16 collects the problems people actually hit, and chapter 17 is a glossary of the terms used throughout.

1.3 Document conventions

Convention	Meaning
BOLD CAPITALS	Something written on the screen: a button, a panel, a knob — MIXER, PANIC, SENS.
Monospace	Something you type, or a name outside choz: a command, a file, a port — <code>choz:midi_in</code> .
F2, Enter, a	A key to press. Capitals mean the shifted key: A is Shift+a.
→	A path through menus: EDIT → Settings → THEME.

Note. Grey boxes like this one carry something worth knowing before it bites: a limit, an interaction with another program, or a setting that only takes effect on the next start.

1.4 System requirements

Requirement	Detail
System	Linux with glibc; x86-64, aarch64 or armv7
Audio	<code>libasound.so.2</code> (required); <code>libjack.so.0</code> (optional, opened at runtime — normally PipeWire's, from <code>pipewire-jack</code>)
Terminal	24-bit colour, at least 80×24 — 120×40 to work comfortably. kitty, Ghostty and WezTerm also draw the wallpaper through the terminal's graphics protocol
Plugin windows	X11 or XWayland, and only to open a plugin's own window: every plugin parameter also appears as a knob in the rack
Privileges	<code>rtprio</code> and <code>memlock</code> (membership of the audio group) recommended for small buffers

1.5 Installing

```
# from a release (no toolchain needed)
tar xzf choz-1.3.16-x86_64-unknown-linux-gnu.tar.gz
cd choz-1.3.16-x86_64-unknown-linux-gnu
./install.sh
```

```
# from the repository
cargo build --release
./packaging/install.sh --binary target/release/choz
```

There are also `.deb` and `.rpm` packages, and an Arch `PKGBUILD`. All of them add a desktop entry under Audio, so choz can be started from the application menu as well as from a shell.

The install leaves four things besides the binary: `choz.clap` (choz's 56 effects plus the arpeggiator, the step sequencer, the metronome and the arranger, as CLAP plugins for other hosts), `choz-rack.clap` (choz itself as a CLAP instrument, section 2.5), the wallpapers, and — when `libpd` is present — the `choz-pd-host` binary that runs Pure Data patches.

Upgrading. `install.sh` lists what it is about to replace — the binaries with their `--version`, and `~/clap/choz.clap` and `~/clap/choz-rack.clap` — and asks before it does. `--yes` skips the question; run without a terminal (a script, CI) it replaces, as it always has.

Packages and old copies in `~/clap`. A CLAP host reads `~/clap` before `/usr/lib/clap`, so a bundle left there by an earlier `install.sh` hides the one a package installs and the host keeps running the old version. The `.deb` asks (debconf) whether to set those copies aside; the `.rpm` and the Arch package cannot ask and set them aside directly. Set aside means renamed to `*.bak`, which no host loads and which can be renamed back.

JACK. Current desktops run PipeWire with its JACK layer (`pipewire-jack`), which is all choz needs for the graph; `jack2` works the same. Neither is required: without `libjack` choz runs on ALSA.

1.6 What an uninstall leaves behind

No uninstaller touches `~/ .local/state/choz`. Projects, plugin paths, scan caches and interface preferences live there and survive upgrades and removals; see section 15 for what each file holds.

Everything else goes. Removing the `.deb`, the `.rpm` or the Arch package removes `/usr/lib/clap/choz.clap` and `/usr/lib/clap/choz-rack.clap` with the program; `install.sh --uninstall` removes the binaries, the desktop files and both bundles from `~/ .clap`, and names each file as it goes.

- Left on purpose: `~/ .local/state/choz` (projects and settings).
- Left because it is yours: any `*.clap.bak` a package set aside in `~/ .clap`. Hosts do not load them; delete them by hand.
- Left by the host, not by choz: its plugin cache. Rescan plugins in the DAW so it forgets choz.

1.7 Starting choz

```
choz                # an empty rack
choz project.yml    # open a project
choz instrument.sf2  # load a file straight into a tab
choz --osc-port 9000 # pin the OSC port instead of letting the OS pick
choz --version       # print the version and exit
choz --help          # print the options and exit
```

1.8 Getting help

HELP → About choz shows the version and the build. The log at `~/ .local/state/choz/choz.log` is the first place to look when something goes wrong, and chapter 16 explains the messages that are worth acting on. Running choz from a terminal also prints its diagnostics to standard error as they happen.

2. First steps

This chapter gets a sound out of choz and back into a DAW. It assumes nothing has been configured.

2.1 Play a SoundFont from a MIDI keyboard

1. Start choz. The rack is empty and says so.
2. Press **F2** to open the **IN** drawer. The note inputs are at the top: every MIDI port on the system, choz's own ports, and OSC.
3. Move to your keyboard's port and press **Enter**. choz makes tab 1 and binds the port to it — the row says → **tab 1**.
4. Press **F2** again to close the drawer, then 1 to open the source picker.
5. Type part of a SoundFont's name, or open the **SF2** tab, and press **Enter**. The bank loads into the tab.
6. Play. If nothing sounds, check the **MONITOR** panel at the bottom: it shows every MIDI message as it arrives, so it answers "is the keyboard reaching choz?" before you look anywhere else.

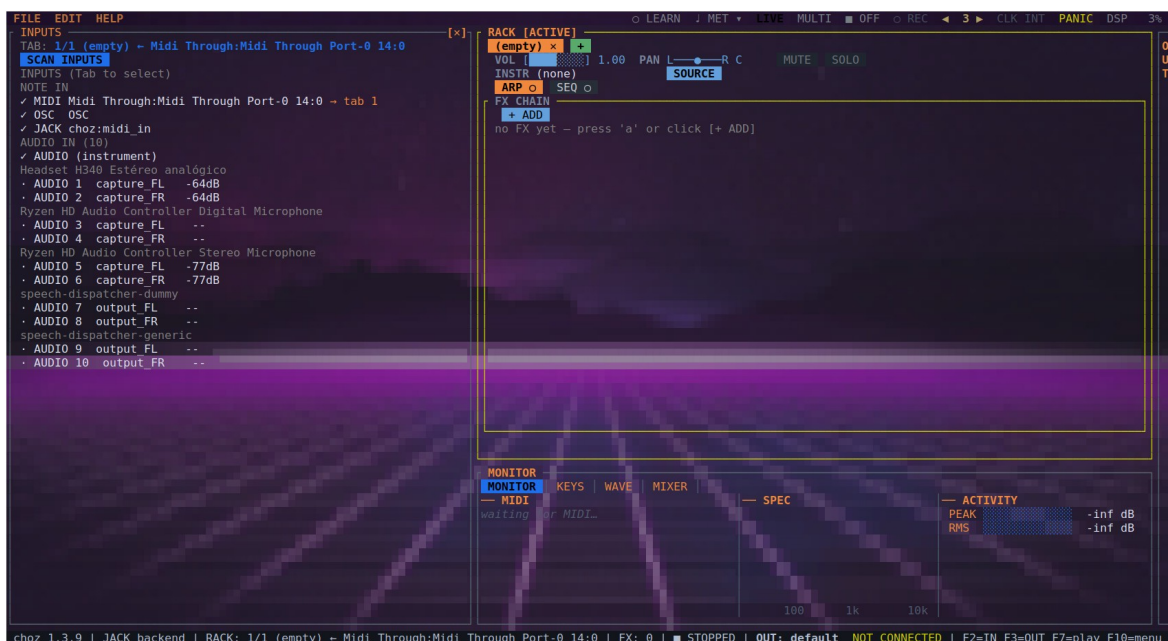


Figure 2.1 — Accepting the MIDI row: choz makes tab 1 and feeds it from that port.

2.2 Add an effect

1. With the tab active, press a to open the effect catalogue.
2. Pick a category on the left — **REVERB**, say — and an effect on the right. **BUILT-IN** at the top narrows the list to choz's own.
3. Press **Enter**. The effect is added to the tab's chain with all of its knobs.
4. Move between knobs with **↑ ↓**, change one with the wheel or w/s, and switch the whole effect on or off with **Space**.

2.3 Play choz from a DAW and record it back

The usual round trip is: the DAW's MIDI out plays a choz tab, and choz's audio comes back into the DAW to be recorded.

1. **MIDI in.** In your DAW's MIDI device list, pick choz MIDI IN as the track's output. choz publishes that ALSA port itself, so nothing else is needed — see section 4.2.
2. **Audio out.** In choz, open EDIT → Settings → **AUDIO** and set the backend to **JACK**, then restart choz. It now has choz:out_1..N on the graph.
3. **Patch.** With any patchbay (qpwgraph, Carla, jack_connect), wire choz:out_1 and choz:out_2 into the DAW's inputs, and arm a track on them.
4. **Per-tab tracks.** To record several tabs separately, give each one its own direct out (section 4.6) and patch those instead of the main mix.

Note. On PipeWire, its JACK layer is enough — there is no need to start a separate jackd. The MIDI half of this round trip does not need the graph at all: choz MIDI IN is an ALSA sequencer port.

2.4 Save the project

Press **F10**, open **FILE** and choose **Save as**. A project is a readable YAML file that carries the rack, the routing, the effect chains, and the plugins' own state — the patch you chose in a plugin's browser comes back with the project.

2.5 Run choz inside a DAW (REAPER)

Section 2.3 runs choz beside the DAW and wires the two together. The other way is to put choz **inside** it: choz-rack.clap is the whole rack as a CLAP instrument on one track — its own window, every tab, and the project saved in the host's session.

REAPER

- Options → Preferences → Plug-ins → CLAP: make sure /usr/lib/clap (packages) or ~/.clap (install.sh) is on the path, and press **Re-scan**.
- Insert **choz (choz)** as a virtual instrument on a track. Its window is choz's interface, keyboard and mouse included.
- The track's MIDI reaches the rack as the host's note port: a tab set to listen to it (section 4.1) plays what the track sends.
- The track's audio arrives as host:in_1 / host:in_2, so a tab with that input processes it: the rack is an effect chain too.
- **Sixteen stereo outputs.** Put a tab on output pair 4 in the rack and it comes out of the plugin's fourth pair. In the FX chain, right-click the plugin → **Build multichannel routing for output of selected FX** to get a track per pair, the way a multi-out sampler is recorded.
- The rack is stored in the REAPER project; reopening the project reopens the rack.

Note. The embedded rack keeps **its own tempo, time signature and play state**: it takes notes and MIDI from the host but not the host's transport. Set choz's tempo (TAP, or the metronome's menu) to the session's. The plugins in choz.clap — effects, arpeggiator, sequencer, metronome and arranger as separate plugins — do follow the host's transport. Only one rack runs per process.

If REAPER shows an old version. A copy of choz-rack.clap or choz.clap in ~/.clap wins over /usr/lib/clap. Remove or rename it, quit REAPER completely (the old one is loaded in memory) and re-scan.

3. choz overview

3.1 The screen

The screen has four fixed areas: the menu bar along the top, the RACK in the middle, the IN and OUT drawers at the left and right edges, and the MONITOR / KEYS / WAVE / MIXER panel along the bottom. The last line is the status line.

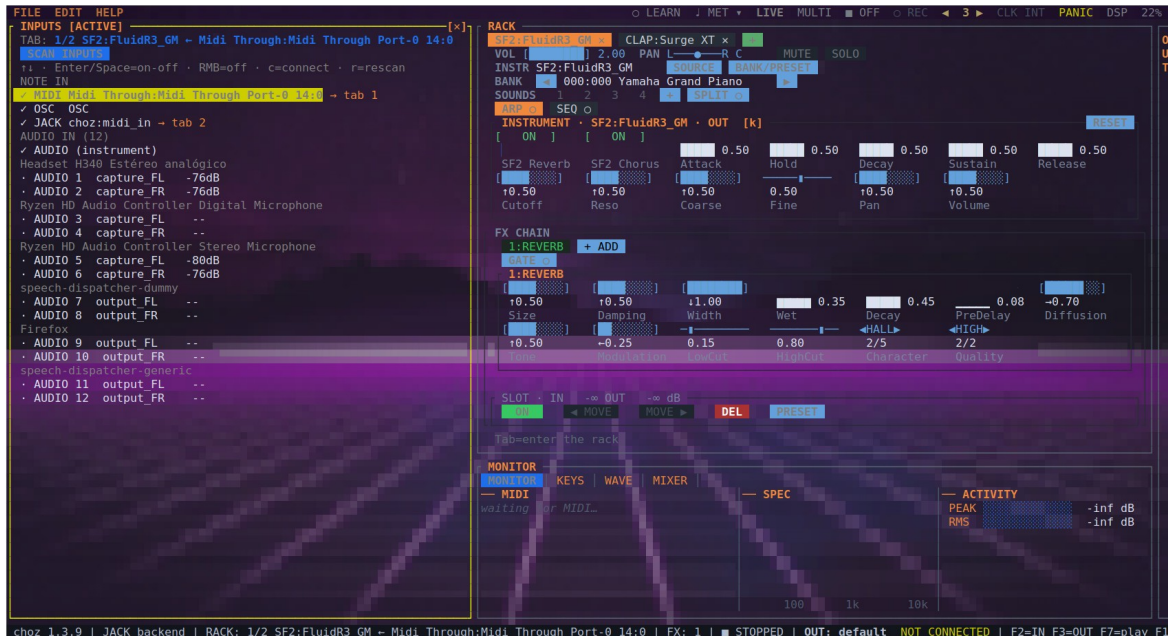


Figure 3.1 — The whole screen: the IN drawer open on the left, tab 1 with a SoundFont and an internal reverb, and the bottom panel below.

3.2 The menu bar and the transport cluster

FILE, EDIT and HELP sit on the left. Everything true of the whole rack rather than of one tab sits on the right:

Control	What it does
LEARN	Arms MIDI learn by pointing: click a knob, then move a controller (section 8.3).
TAP	Plays the tempo in. Four clicks set it, and the fourth starts the metronome (section 10.3).
MET and ▼	The metronome and its options — click, tempo, time signature, grouping, sound, output, level.
LIVE / MULTI	Whether every tab plays at once or each listens on its own MIDI channel (section 8.1).
▶ ON / ■ OFF	The transport: what starts every sequencer, every synced delay and the automation.
REC	Arms automation recording. A right-click clears the lanes.
◀ n ▶	The automation loop, in bars. Click the left half to shorten it, the right half to lengthen it.
CLK	The clock source: choz's own, or the external device driving it, by name.
PANIC	Silences every note that is still sounding.
DSP	What the audio thread is spending, as a percentage of the block budget.
MEM	The memory choz holds right now, resident. It turns amber when the machine has less than 15% left — where a streamed sample library starts reading back late. The first thing given up on a narrow window.

When the window is too narrow for the whole cluster, controls are given up in a fixed order: the ones a key can still reach go first, and the switches with no other home go last.

3.3 The rack

One tab per channel. A tab holds its input binding, its instrument, its effect chain, its arpeggiator and sequencer, its volume, pan and routing. [and] move between tabs; the mouse works everywhere.

3.4 The drawers

F2 opens IN, **F3** opens OUT. They are lists of everything choz can hear and everything it can reach, and binding is done from them (chapter 4).

3.5 The bottom panel

F5 cycles MONITOR, KEYS, WAVE and MIXER; clicking a tab goes straight there (section 11.4).

3.6 The status line

Version, audio backend, active tab, how many effects it carries, the transport, and where the output is going.

3.7 Mouse, wheel and keyboard

Everything on the screen is clickable, and the wheel over a knob changes its value. The keyboard reaches everything the mouse does: chapter 14 is the full list. Inside the rack, **k** moves the cursor between the instrument and the effects, the arrows walk the knobs, and **Enter** opens a list for a knob that has named values instead of a range.

4. Inputs and outputs

4.1 The IN drawer

At the top are the note inputs: every MIDI port on the system, OSC, choz's own JACK port (choz:midi_in) and choz's own ALSA port (choz MIDI IN). Below them are the audio inputs the device or the graph offers, with their level in dB live, grouped by device — and, on PipeWire, the outputs of other applications as well.

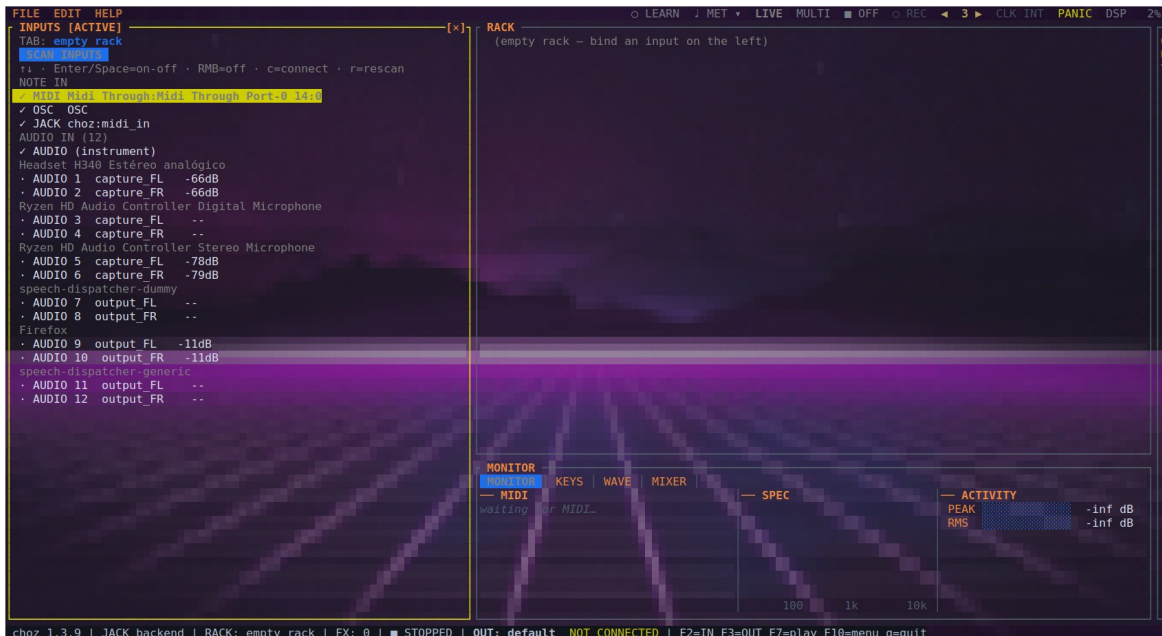


Figure 4.1 — The IN drawer: note inputs and the audio inputs found, each with its current level.

Key	Action
Enter / Space	Bind that input to a tab — choz makes one if there is none. Again to unbind (a right-click does the same).
c	Connect or disconnect the port.
r	Rescan the inputs.

choz connects to every enabled MIDI input by itself, so a controller plugged in mid-session starts playing without any wiring. Ports you never want are switched off in this drawer and stay off.

4.2 choz MIDI IN

ALSA's sequencer only lets a program *subscribe to* ports that already exist, and a DAW's MIDI output is not one of them: it opens the devices it was told to and publishes nothing anybody else can send to. So choz publishes the port instead. choz MIDI IN is a writable port on the ALSA client choz: a DAW picks it out of its own MIDI output list and plays a rack tab — with no snd-virmidi, no a2jmidid and no JACK graph.

- It is opened once and lives as long as choz does. Plugging or unplugging a controller re-scans the inputs, and a port remade there would drop the DAW's subscription every time somebody touched a USB cable — silently, because neither side reports it.
- It is the first row of the note inputs, and it can be switched off like any other port. Switching it off closes it, so it leaves the DAW's list too.
- It carries everything a DIN cable carries: notes, controllers, program change, pitch bend and MIDI Clock — so a DAW's transport can drive choz's tempo (section 8.6).

4.3 choz on the JACK/PipeWire graph

With the **JACK** backend (section 13.1), choz registers as the client choz:

Port	What it is
choz:out_1 ... out_N	The output channels: the device's own, plus two more for each tab that has a direct out.
choz:in_1 ... in_N	The capture jacks in the graph, whatever card they belong to.
choz:midi_in	MIDI from the graph: notes, controllers and clock published by a DAW on the same session.
choz:midi_out	What the arpeggiator, the sequencer and the tabs send out — to a synth on the graph, or to the DAW's recorder.

Nothing is auto-connected to the two MIDI ports: what talks to choz is a patch you make, not a guess choz makes for you.

4.4 The OUT drawer

F3 lists the playback device and its channels, and the MIDI output ports available — including choz:midi_out.

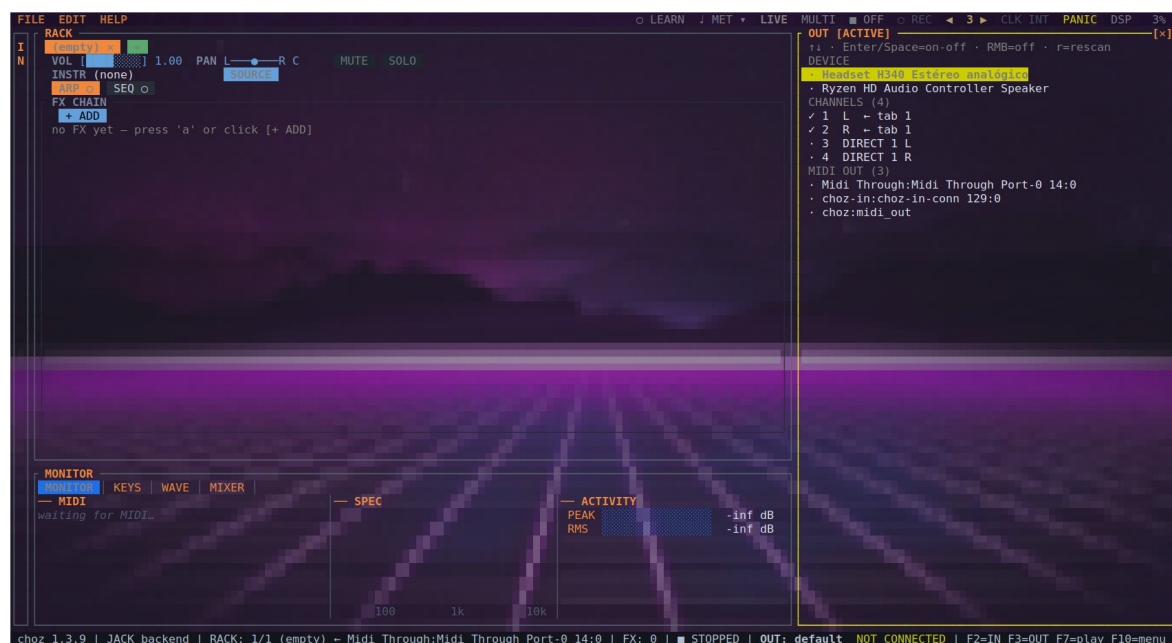


Figure 4.2 — The OUT drawer: output device, channels, and MIDI output ports.

4.5 Audio inputs and the microphone

An audio input binds to a tab exactly like a note input: open IN, pick the channel, press Enter. The tab gains two controls of its own — **IN**, the input trim (< and >), and **SENS**, the sensitivity of the audio-to-note follower (; and :) — and the **A → M** button, which turns what comes in into MIDI notes for the instrument or for other tabs.

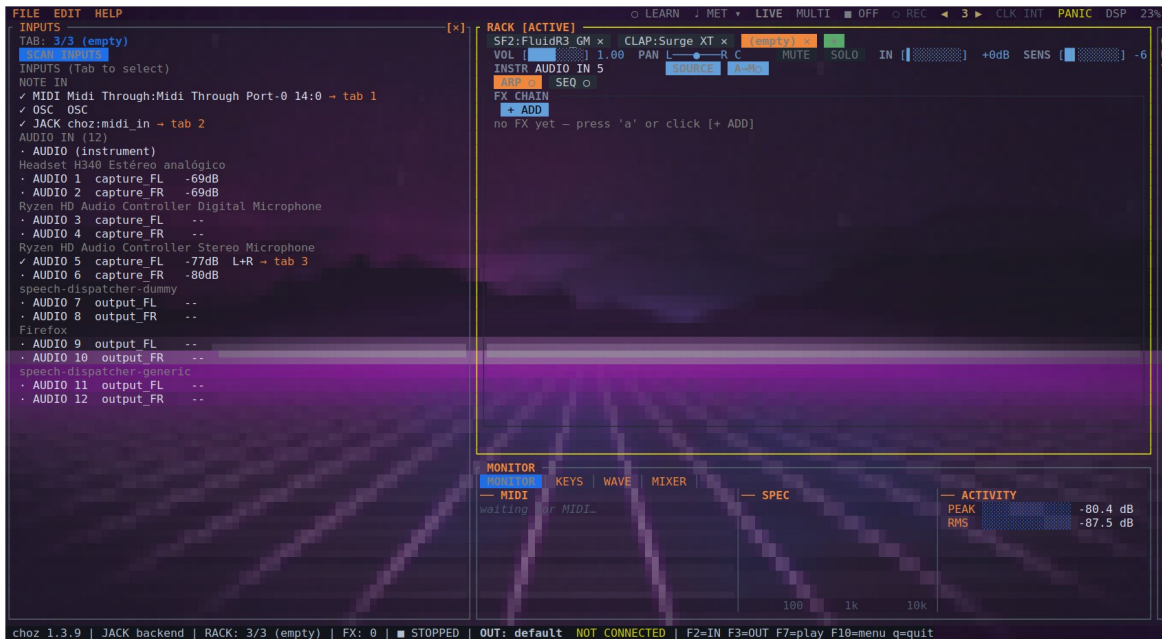


Figure 4.3 — A stereo microphone bound to tab 3: the IN row with the trim, SENS, and the A → M button.

- A mono channel stays mono all the way through — one fader on the mixer, one port on the graph — until something in the chain has a reason to pull the two sides apart.
- n levels the tab against what it has played since it was loaded; N levels the whole rack.
- The feedback guard (section 13.1) cuts the loop when a microphone starts to howl.

4.6 Direct outs

See section 11.3: a tab can leave the master mix through a pair of ports of its own, which is how a DAW records each tab on its own track.

5. Instruments

5.1 The source picker

With a tab active, 1 (or i) opens the picker. The tabs across the top filter by format — ALL, CLAP, SF2, WAV, SFZ, LV2, VST2, VST3, DSSI, LADSPA. Type to search, move with the arrows, accept with Enter; **F5** rescans.

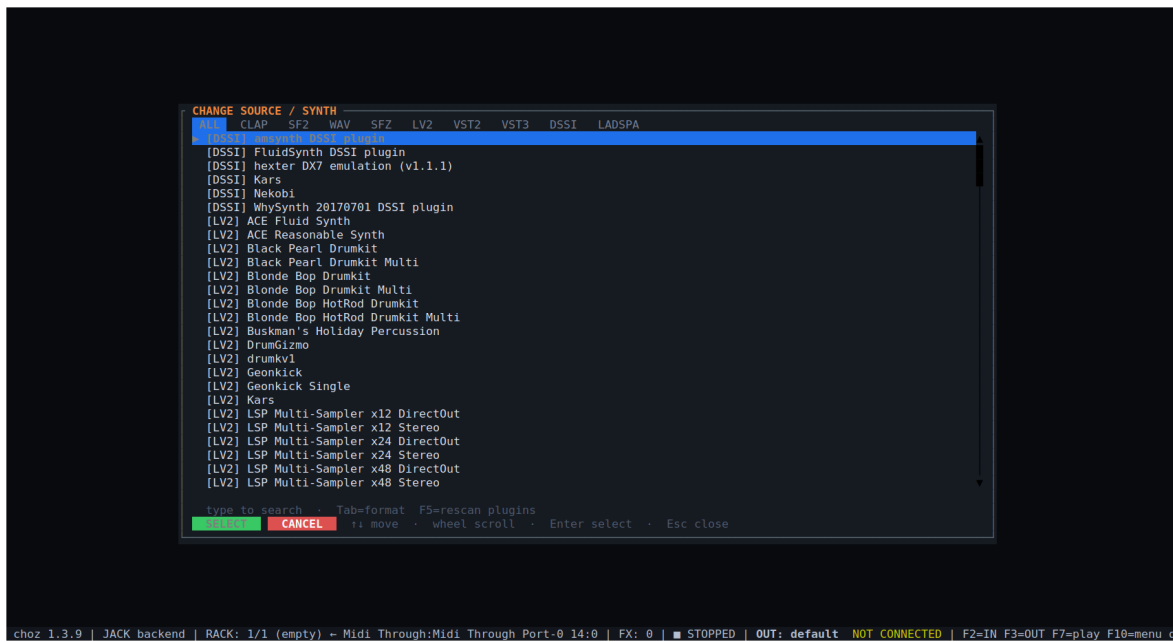


Figure 5.1 — The source picker: every instrument found, with its format in brackets.

5.2 SoundFonts

choz synthesises SF2 with its own engine, with no dependency on FluidSynth. Loaded, the tab shows the preset on the **BANK** row and the engine's knob box: the bank's reverb and chorus sends, the envelope (Attack, Hold, Decay, Sustain, Release), the filter (Cutoff, Reso), tuning (Coarse, Fine) and output (Pan, Volume).

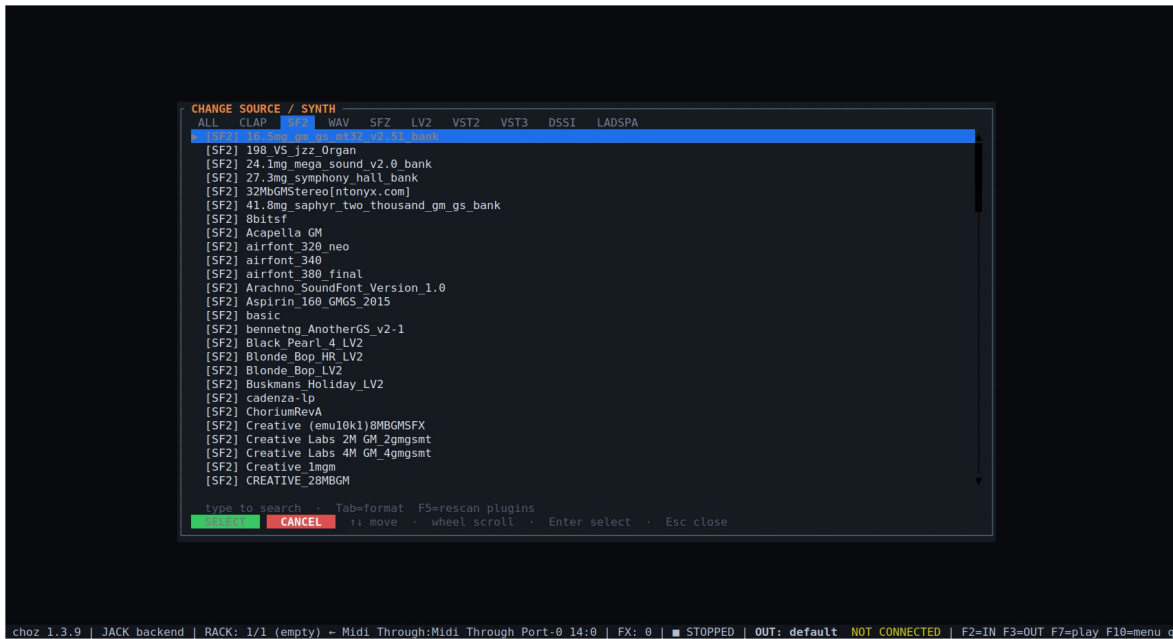


Figure 5.2 — The picker's SF2 tab: the SoundFont banks installed on this machine.



Figure 5.3 — A SoundFont loaded in tab 1: the BANK/PRESET row and the SF2 engine's knob box.

5.3 SFZ and WAV

SFZ instruments play on choz's own 32-voice sampler and a WAV plays as a rack source; both load from the same picker. A SoundFont player installed as a plugin appears under its own format tab and is used like any other plugin.

5.4 Banks and presets

2 (or b) opens the bank and preset browser. The highlighted row sounds while you move through it, and **SELECT** keeps it. When a plugin's patches are files rather than numbered programs, the same button opens the categories its own window would show.

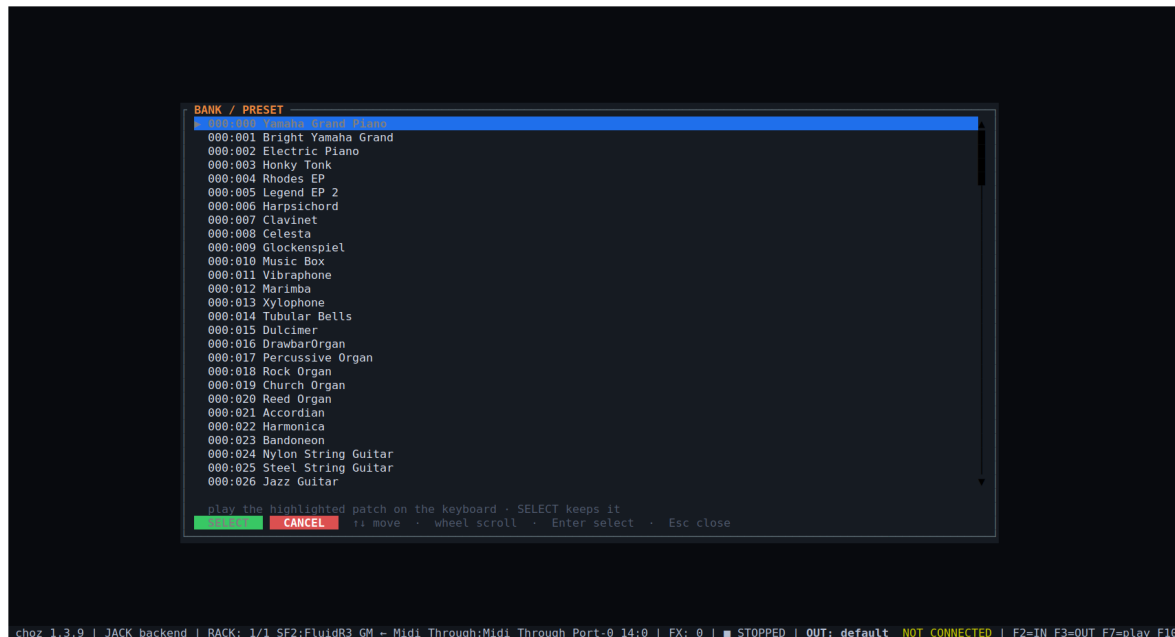


Figure 5.4 — An SF2 bank browser: the 128 programs of bank 0 and the file's other banks.

5.5 Parameters, pages and search

p opens the parameter list with exact values; l on a row learns a CC for it. A plugin with hundreds of parameters is shown two rows at a time — **PgUp** and **PgDn** turn the page, and learned CCs move with the box, so eight faders reach every parameter the plugin has. / searches a knob by name and jumps to the page holding it.

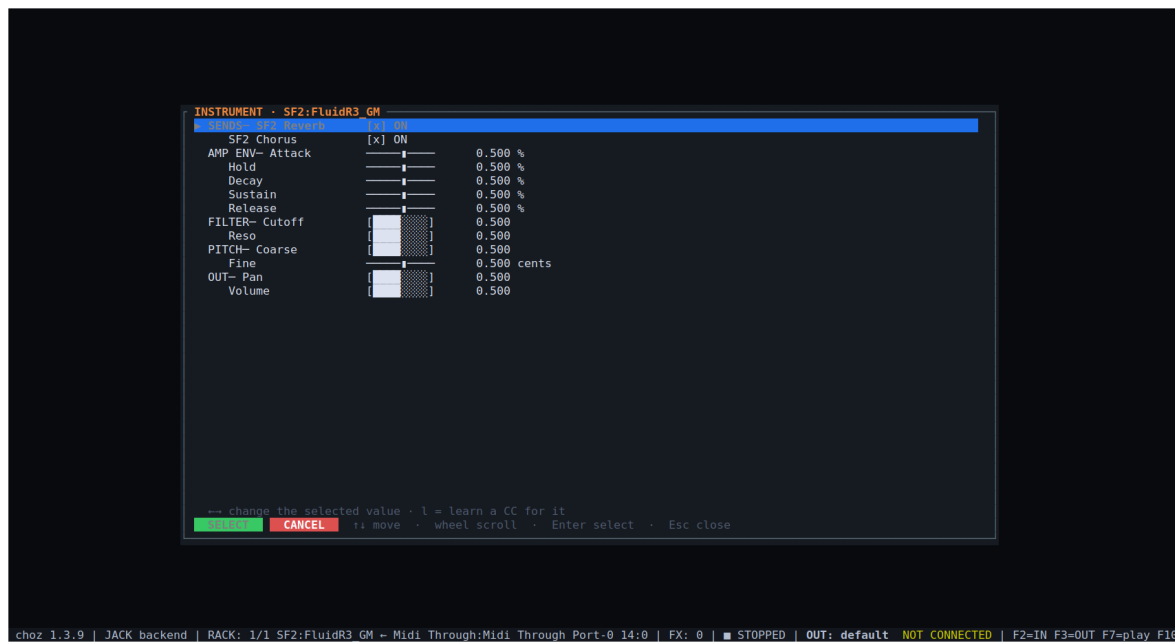


Figure 5.5 — A SoundFont's parameters: sends, envelope, filter, tuning and output.

5.6 The sampler

choz plays folders of samples as instruments. **choz-sampler** is always in the source picker under **CLAP**; sample libraries found on the plugin paths (~ / Samples and ~ / samples by default, Preferences → Plugin paths) are listed under the **SAMPLE** chip. Picking the sampler puts it in the tab empty; **LOAD** asks for a folder, a .zip or a single audio file (.wav, .flac, .aiff, .mp3, .ogg).

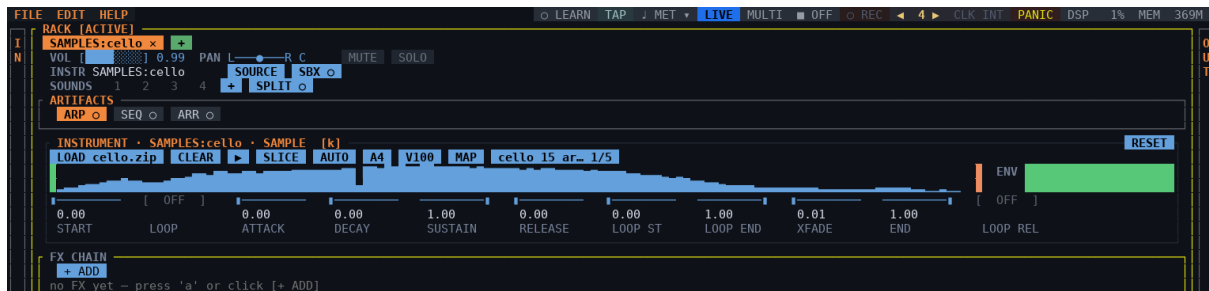


Figure 5.6 — The sampler playing Philharmonia's cello: the note's own waveform with its start mark, the envelope, and the knobs.

How a folder becomes an instrument. File names first — violin_C4_mf_RR2.wav says the note, the dynamic and the take — then the folder names, then, only for what is still open, the audio itself (pitch detection). Dynamics become velocity layers, takes become round robins, and a pack with several articulations keeps all of them, selected by **keyswitches** on the keys below the playing range. Each instrument is levelled on loading, so a flute and a tuba start at the same height, and large libraries stream from disk instead of loading whole into memory.

The panel. A tab with the sampler shows its own panel in the RACK:

- **LOAD** (with the folder it is playing) and **CLEAR**.
- The layout: **AUTO** (worked out from the files), **STRETCH** (one sample across the keyboard), **KIT** (one sample a key, drums on their GM keys) and **SLICE**, which cuts the longest file into 8, 16, 24 or 32 pieces, one a key from C2.
- ▶ / ■ to audition without a keyboard, with the key (**NOTE**) and the velocity (**V**) it plays, and the articulation when the pack has several.
- **MAP** opens the mapping editor: a row a region with root, low key, high key and tune. Arrows move, PageUp/PageDown move by an octave, F2 saves an .smpreset beside the folder and reloads the tab from it, F3 relinks a library that moved, Esc discards.
- The waveform of the sample the key plays, with the start point and the slices marked — a click on it sets the start — and the envelope, which can be dragged.

Its knobs are ordinary instrument parameters — **START**, **LOOP**, **ATTACK**, **DECAY**, **SUSTAIN**, **RELEASE**, **LOOP ST**, **LOOP END** and **XFADE** — so they are saved with the project, move with the arrows and can be MIDI-learned. The loop runs while the key is down and is crossfaded at the join. Channel and polyphonic aftertouch arrive as expression (CC 11).

6. Effects

6.1 The catalogue

a (or [+ **ADD**] in the FX CHAIN section) opens it. The left column is categories with how many effects each holds; the right column is the list. The tabs filter by format, and **BUILT-IN** leaves only choz's own.

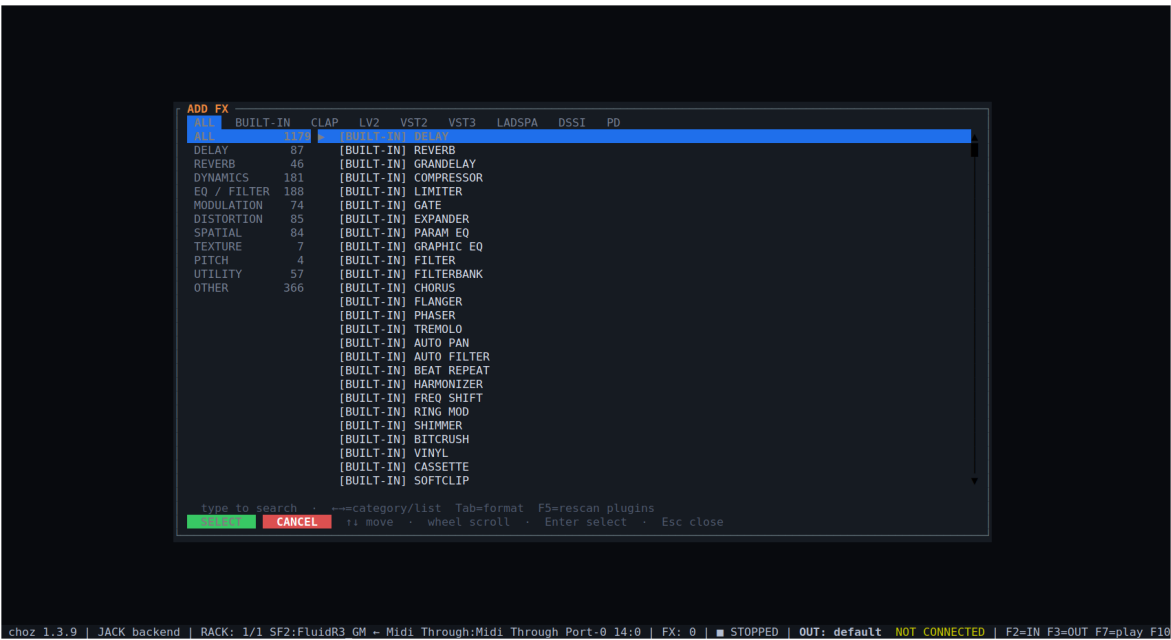


Figure 6.1 — The effect catalogue: everything available between the built-ins and the installed plugins.

6.2 The chain

A tab holds up to twelve effects, in order. Each shows all of its knobs and, underneath, its slot row: ON/OFF, input and output level in dB, moving it along the chain, deleting it, and its preset buttons.

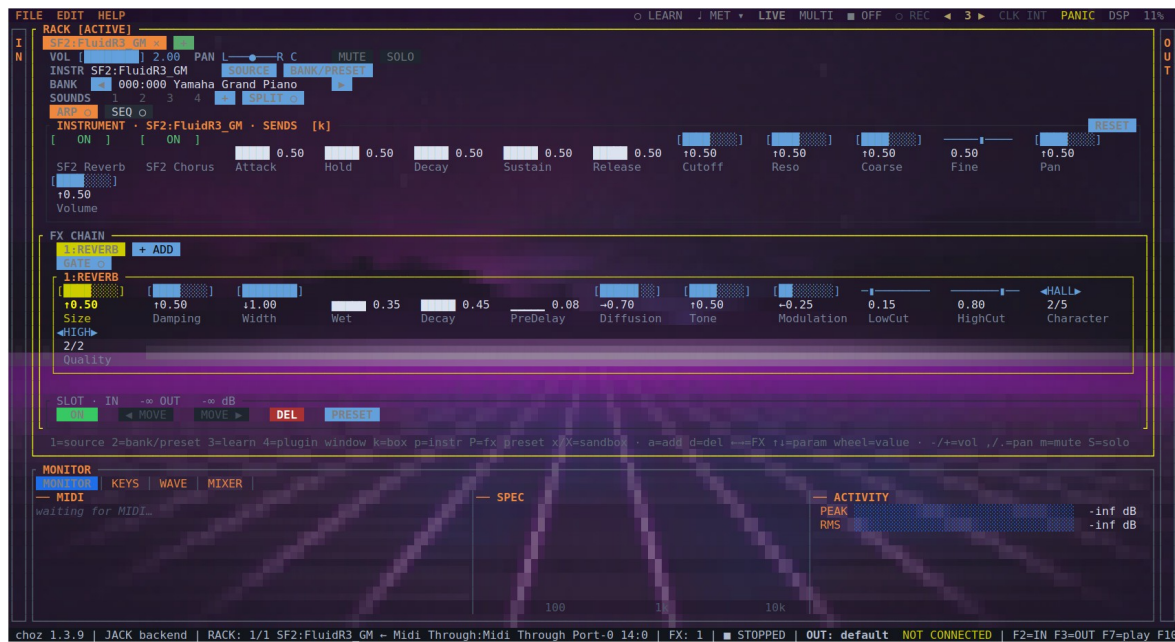


Figure 6.2 — The internal reverb in the chain, with its own parameters.

Key	Action
Space	Switch the selected effect on or off
← →	Move between effects in the chain
↑ ↓, wheel, w/s	Pick a knob and move it
d	Delete the selected effect
G	Open the effect's own window, if it is a plugin that has one
X	Run the effect isolated in another process

6.3 The built-in effects

Fifty-six effects ship with choz. They need no plugin, they are always present, and they are the ones published in `choz.clap` for other hosts.

Category	Effects
Delay (5)	DELAY, GRANDELAY, SPACE ECHO, REVERSE, MULTI-TAP
Reverb (3)	REVERB, SHIMMER, PLATE
Dynamics (9)	COMPRESSOR, LIMITER, GATE, EXPANDER, SIDECHAIN, DE-ESSER, TRANSIENT, MULTIBAND, ENVELOPE
EQ / Filter (6)	PARAM EQ, GRAPHIC EQ, FILTER, FILTERBANK, ISOLATOR, LADDER
Modulation (9)	CHORUS, FLANGER, PHASER, TREMOLO, AUTO PAN, AUTO FILTER, FREQ SHIFT, RING MOD, VIBRATO
Distortion (11)	BITCRUSH, VINYL, CASSETTE, SOFTCLIP, SATURATOR, WAVESHAPER, TUBE SAT, AMBER FANG, VELVET FUZZ, EXCITER, BASS ENH
Spatial (2)	WIDENER, PAN
Texture (4)	BEAT REPEAT, LOOPER, PROTOCOSMOS, Z5 TEXTURE
Pitch (4)	HARMONIZER, VOCODER, AUTO-TUNE, PITCH SHIFT
Utility (3)	GAIN, PHASE INV, MONO

6.4 Effect presets

Each slot can save and load a preset of its own effect, so a compressor setting that works on a voice can be carried to another tab or another project without saving the whole rack.

6.5 Chord source and the gate

- `c` chooses which keyboard the selected effect takes its chord from — the harmoniser, for instance, with two controllers on a hub.
- `c` opens the gate: another tab drives this effect's dry/wet or one of its knobs, with its level or with the notes it receives. That is how a kick ducks a pad, or a vocal opens a delay.

7. Plugins

7.1 Formats

choz hosts CLAP, LV2, VST2, VST3, LADSPA, DSSI and Pure Data patches, as instruments and as effects. What appears in the rack is what the plugin declares: a parameter that says it cannot be automated is not drawn as a knob, and a parameter whose whole integer range reads as two words is drawn as a switch rather than a fader.

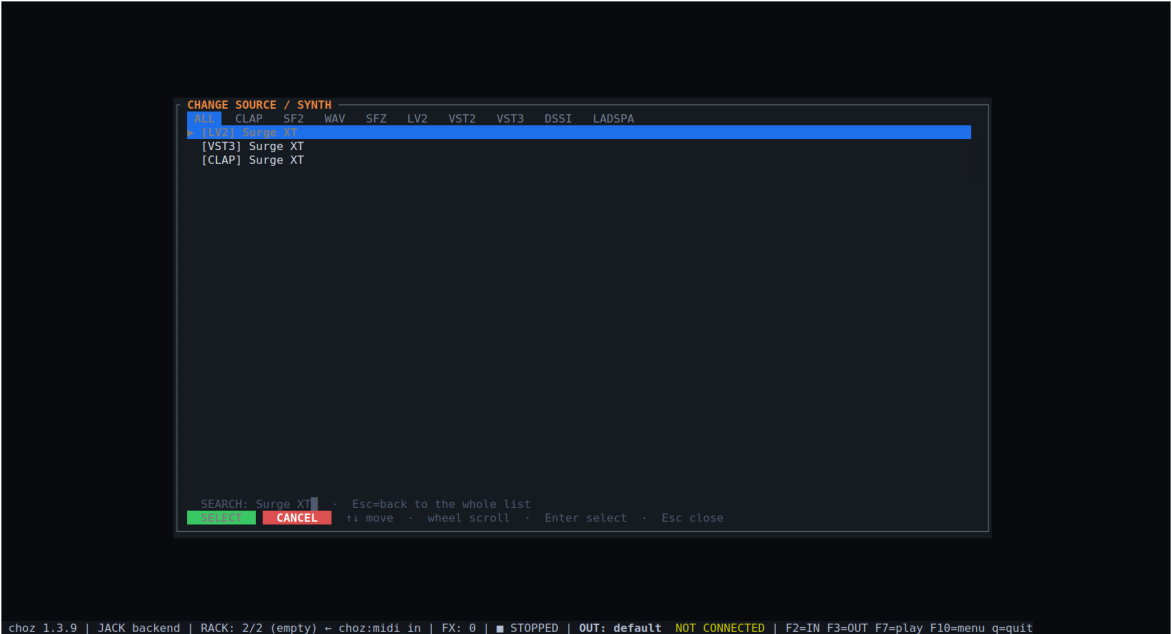


Figure 7.1 — One synthesiser found in three formats; which you pick only changes the host choz uses.



Figure 7.2 — A plugin loaded: its macros, its sends, and the first page of its parameters.

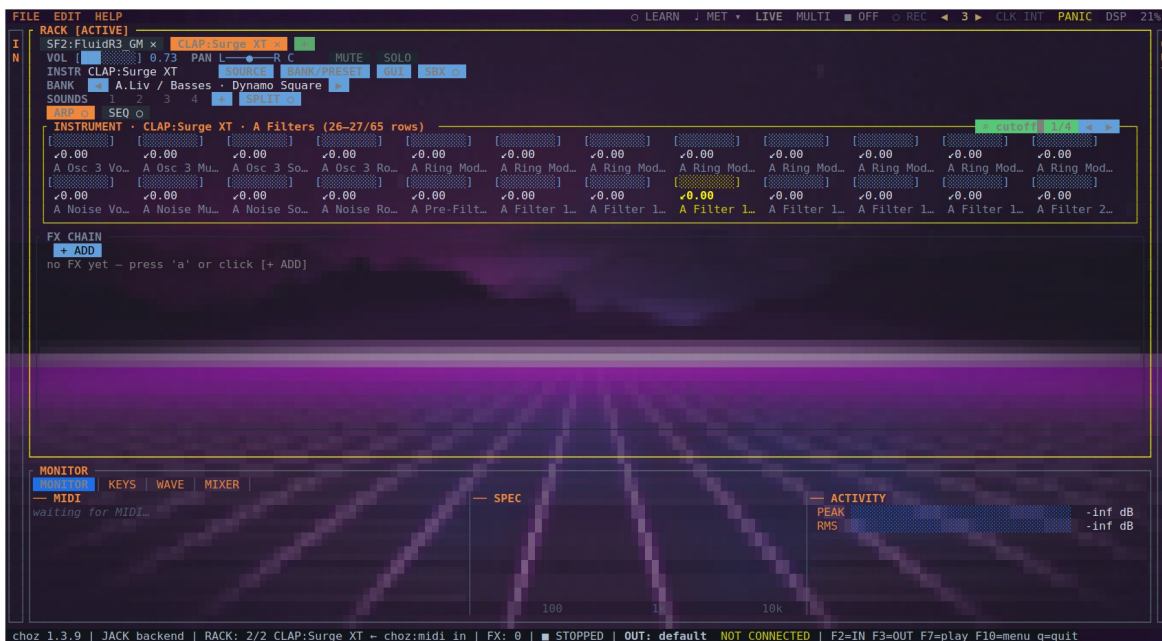


Figure 7.3 — Searching a knob by name jumps to the page that holds it and says how many matches there are.

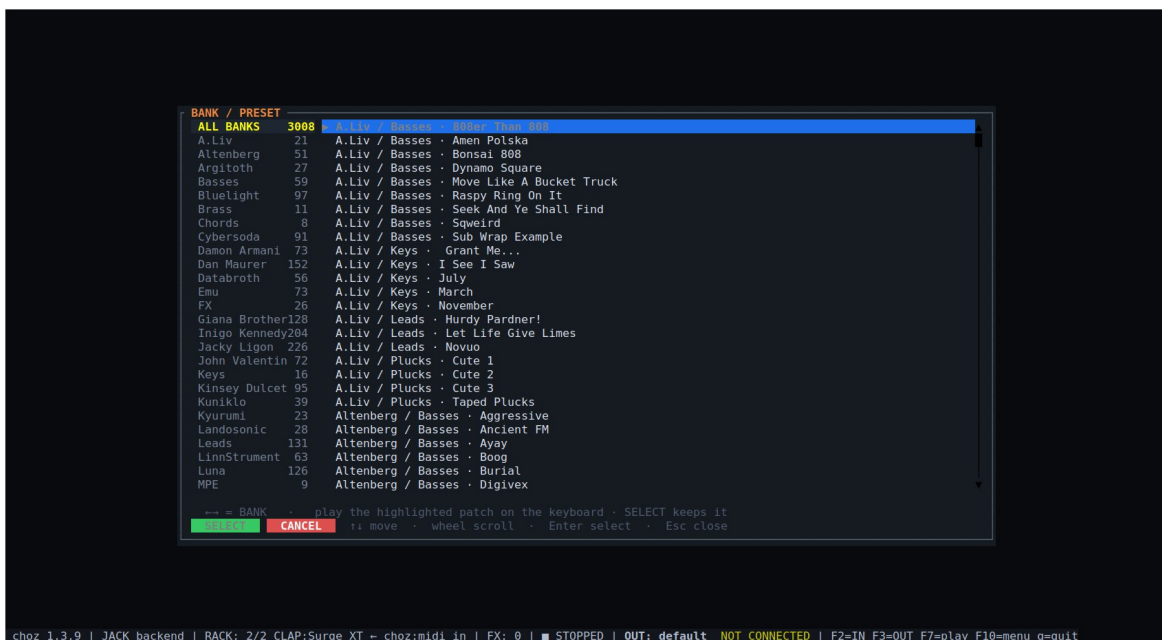


Figure 7.4 — A plugin whose patches are files: its own categories, browsed from choz.



Figure 7.5 — An LV2 synthesiser in the rack, its parameters as native choz knobs.

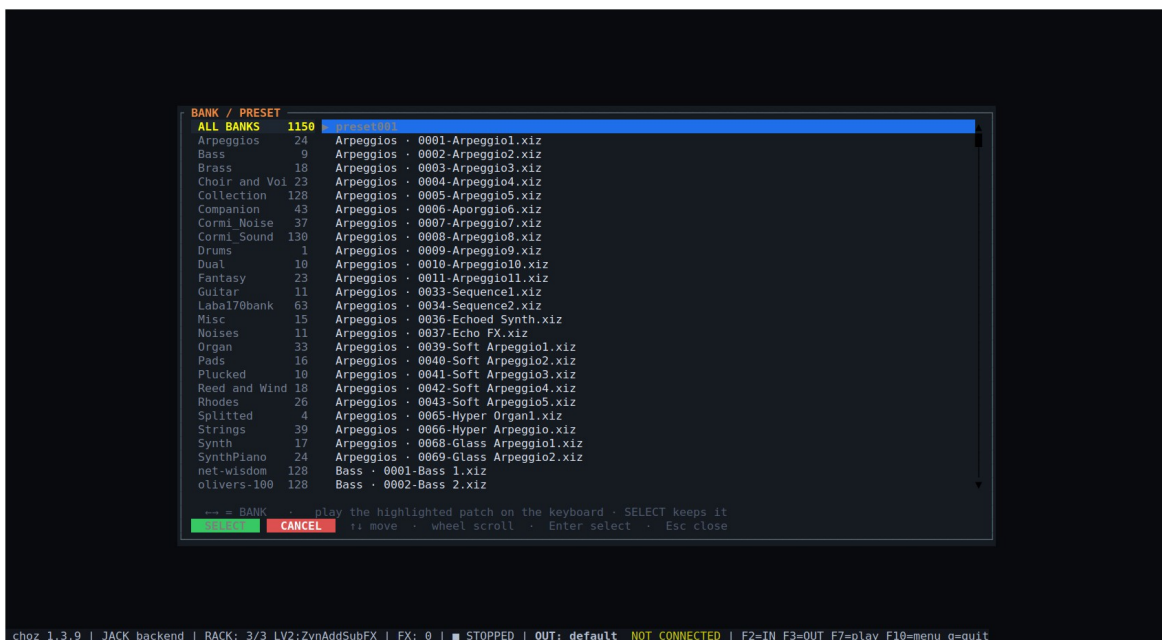


Figure 7.6 — The same plugin's presets, grouped by folder.

7.2 Scanning and paths

Plugins are scanned in the background and cached, so the picker opens instantly on later starts. EDIT → Rescan plugin paths repeats the scan; F5 in the picker does the same. Where choz looks is set in Settings (section 13.2).

7.3 A plugin's own window

g opens the instrument's window and G the selected effect's. choz embeds it in an X11 window of its own — LV2 ui:X11UI, VST2 effEditOpen, CLAP clap.gui on the host's timer, VST3 IPlugView on the Linux event loop. LADSPA, DSSI and Pure Data have no embeddable window; their parameters are driven from the rack.

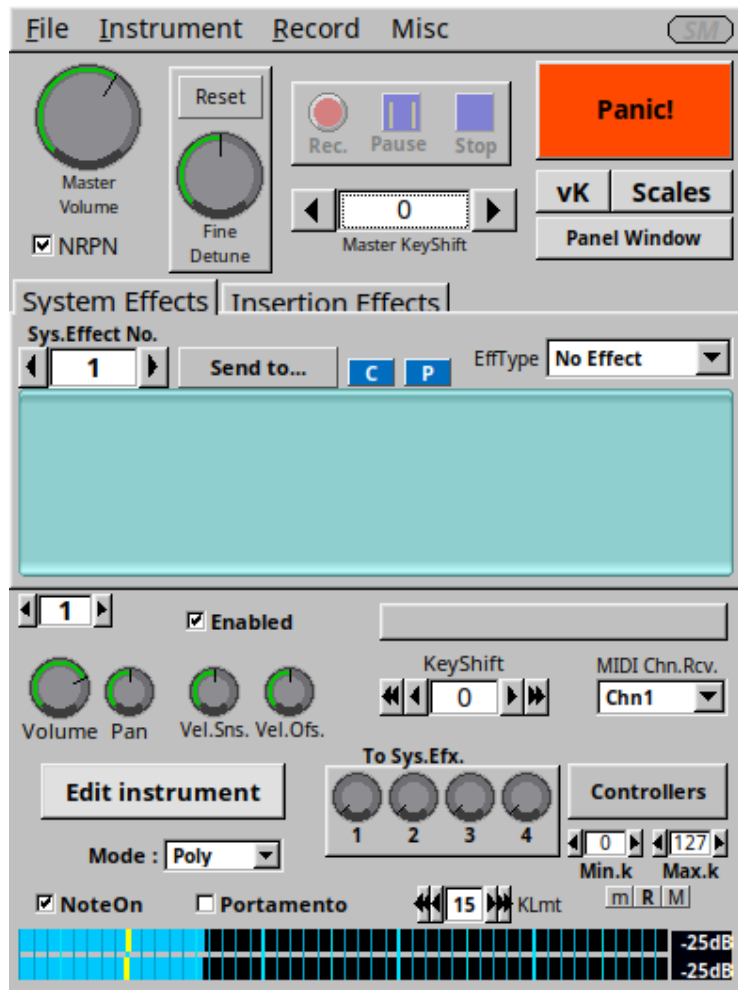


Figure 7.7 — A plugin's own window, opened from choz with g.

Moves made inside that window are followed by choz, so "move the knob in the plugin, then move a fader on the controller" is enough to bind a CC to it.

An instrument that publishes its oscillator's harmonics — over its own OSC server, say — also gets the **HARMONICS** view (h, or the **HARM** button), which draws the harmonic series as bars and edits it in place.

7.4 Sandbox and quarantine

x runs the instrument in a child process and X the selected effect (the **SBX** button does the same). Audio and the window travel over shared memory, so a plugin that crashes does not take choz with it. Before trusting a new plugin, choz probes it in a separate process — a plugin that dies there is remembered and hosted out of process from then on.

The **SBX** button also reads what the sandboxed plugin has cost: a number and lost counts audio blocks the child failed to answer in time (silence the user heard), and a number with v counts times it crashed and was restarted. A plugin that keeps crashing — five times inside ten seconds — or that floods its own log — more than 16 MiB of output in two seconds, the case a badly behaved LV2 synth caused in practice — is no longer restarted: choz gives up on it, quarantines it for next time, and the button reads DEAD instead of counting another restart.

Note. CHOZ_SANDBOX_GUI=1 isolates every plugin with a window up front, which is the setting to reach for on a machine where one badly behaved GUI keeps taking the rack down.

7.5 Plugin state in a project

A project saves what a parameter list cannot: VST2 chunks, VST3 IComponent::getState, clap.state and LV2 state#interface — the patch chosen in the plugin's own browser comes back with the project.

8. Playing and MIDI

8.1 LIVE and MULTI

In **LIVE**, every tab plays at once. In **MULTI (F4)**, each tab listens on a MIDI channel of its own, shown as **CH n** beside the instrument's name — a keyboard split across a rack, or a DAW driving eight tabs down one port.

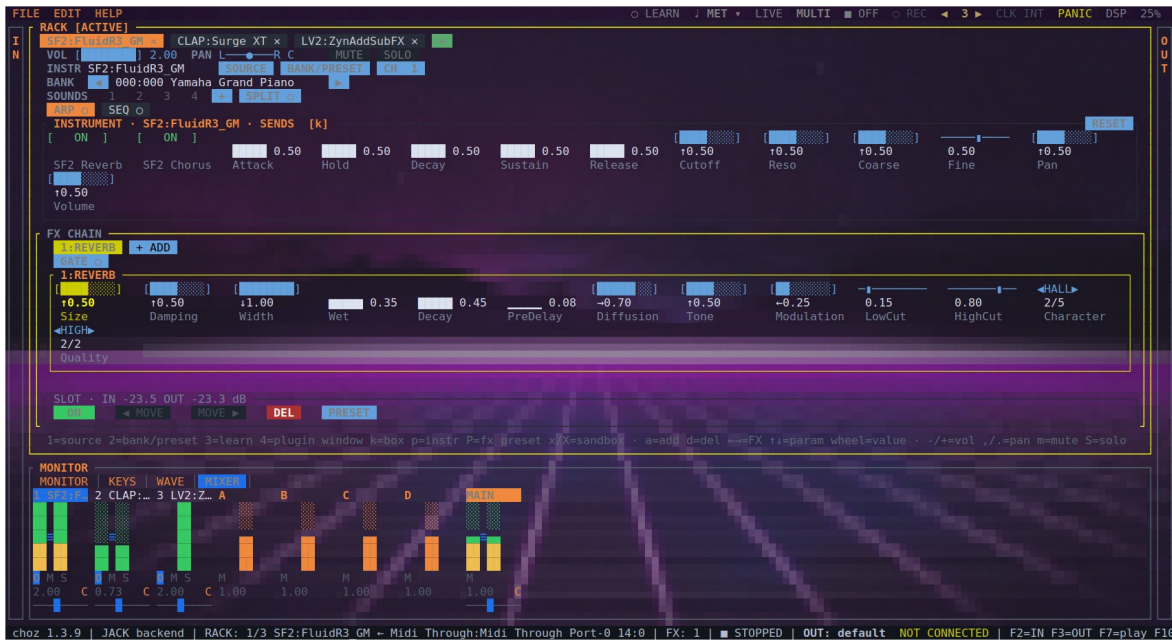


Figure 8.1 — **MULTI** mode: each tab bound to its own MIDI channel.

8.2 The monitor

The **MONITOR** panel shows incoming MIDI as it arrives — port, message, note, velocity — beside the spectrum of the output and the peak and RMS meters. It answers three questions at once: did it arrive, what came out, how loud.



Figure 8.2 — **MONITOR**: **NOTE ON / NOTE OFF** with velocity, the spectrum, and the meters.

8.3 MIDI learn

1 lists the learnable targets of the active tab: volume and pan, mute and solo, bank up and down, parameter page, instrument reset, effects on and moved, add effect, the arpeggiator (on/off, tap, hold), the four sounds, the tabs, and every instrument parameter. Pick one, then move the controller. 3 does it the other way round — pick the control on screen first — and the **LEARN** button lets you learn by clicking a knob.

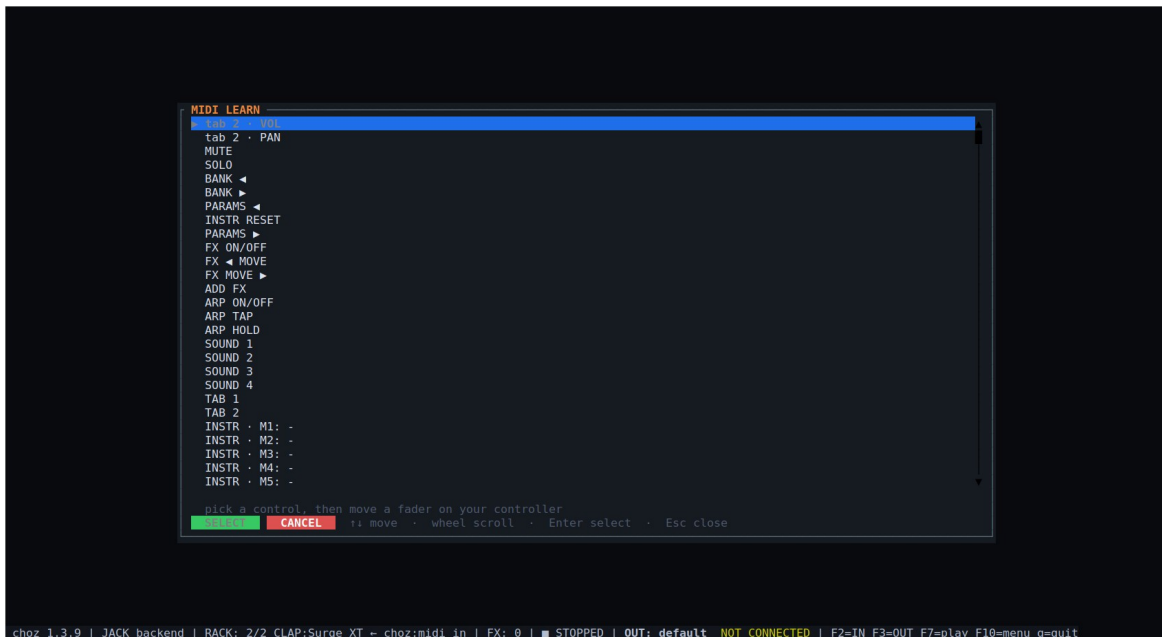


Figure 8.3 — MIDI learn: the tab's targets, including every plugin parameter.

8.4 KEYS and panic

The **KEYS** panel draws the keyboard with the notes sounding, the pitch bend and the modulation wheel, one colour per tab. P (or **F9**, or **PANIC**) silences everything left hanging.

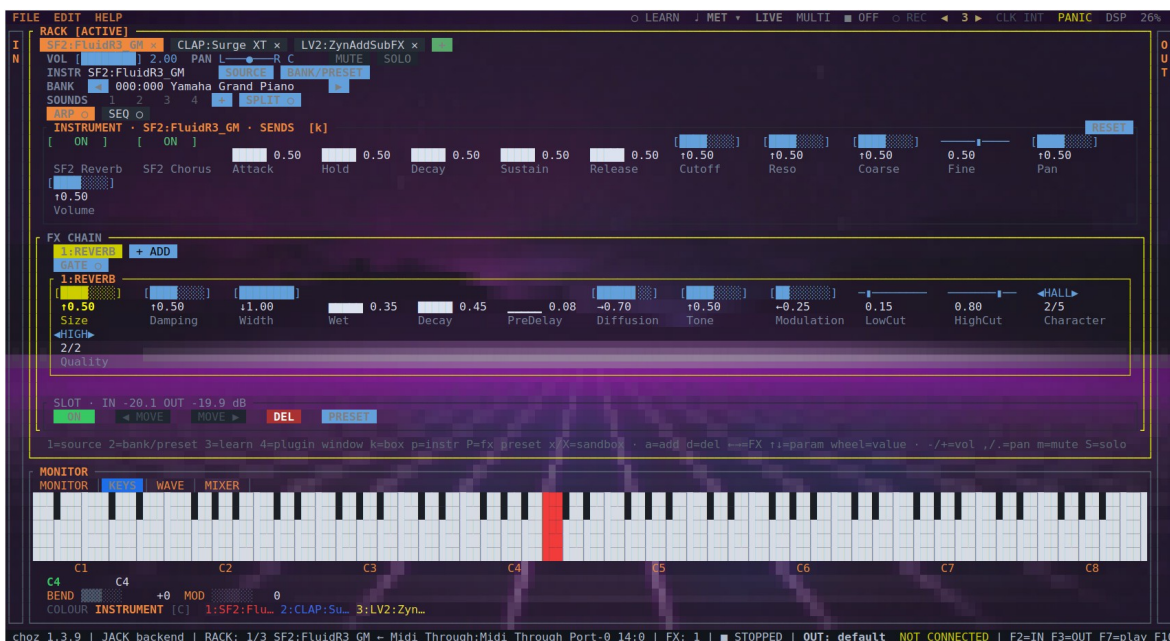


Figure 8.4 — The KEYS panel: a note sounding, bend and modulation, and each tab's colour.

8.5 OSC

choz listens for OSC as one more note input; `--osc-port` pins the port instead of letting the OS pick one. It appears in the IN drawer and binds to a tab like any other source.

8.6 Clock

CLK on the menu bar chooses what drives the transport: choz's own clock, or an external device by name. MIDI Clock arriving on a port choz is following moves the tempo, and **START**, **CONTINUE** and **STOP** move the transport with it — so a DAW's play button starts the arpeggiators, the sequencers and the synced delays.

9. Arpeggiator, step sequencer and arranger

9.1 Arpeggiator

A switches on the active tab's arpeggiator (so does the **ARP** button). Its controls are the mode (UP, DOWN and the rest), the division (1/16, 1/8 ...), BPM, gate, swing, octaves, **HOLD**, **CHORD** and **TAP**. HOLD behaves like a hardware sequencer's: letting go keeps the chord sounding, and the next key pressed with nothing held starts a new chord instead of adding to the old one.

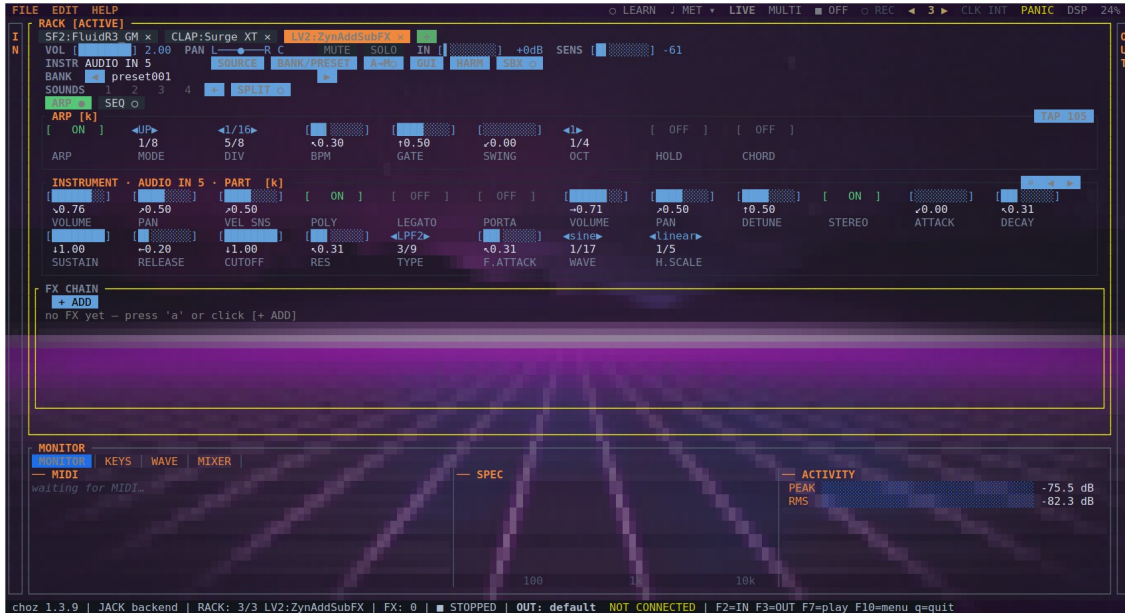


Figure 9.1 — The tab's arpeggiator: mode, division, BPM, gate, swing, octaves, hold, chord and tap.

One clock. The arpeggiator, the sequencer, the arranger and the metronome count the same position — the transport while it rolls, a free clock that never stops otherwise — so anything started at any time lands on the same grid. The first note of a chord sounds when the key is pressed; the rest fall on the grid. Swing lengthens the same steps it does in the sequencer.

9.2 Step sequencer

Every tab also carries a sequencer built like a hardware step machine: eight tracks (A–H), sixteen steps, eight parts and a song chain. It is drawn above the instrument because that is the order the notes travel in. **REC** writes what you play, quantised to the step the play head is on, and each step passes through the tab's arpeggiator when there is one.

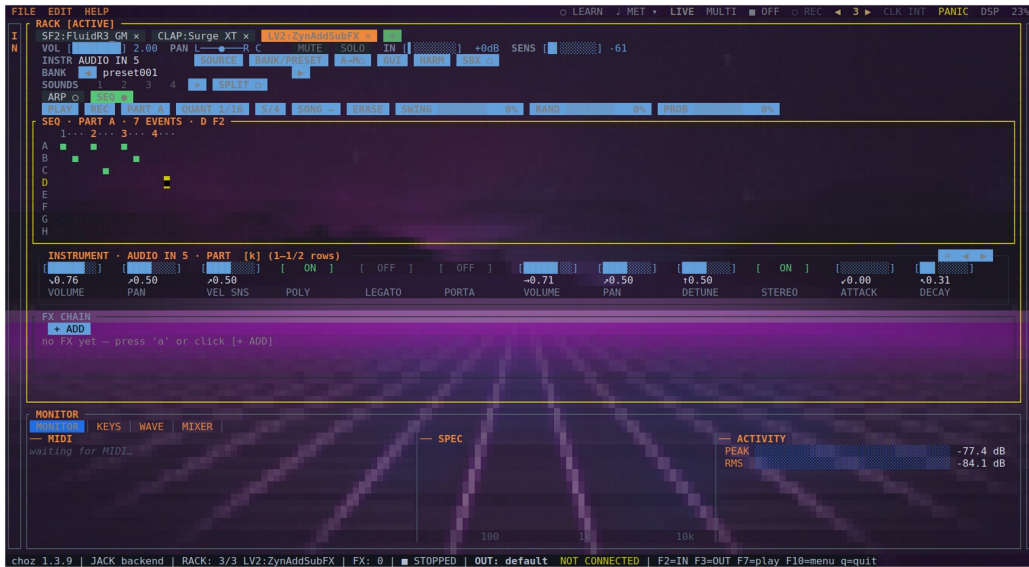


Figure 9.2 — The sequencer: play, record, part, quantise, time signature, song chain, clear, swing, randomness and probability.

-  plays the pattern from the top of the song chain,  leaves and comes back in the same part of the chain,  stops and rewinds. **K** in the box still toggles play and stop.
-  **takes the metronome with it** when something was playing: a click over a stopped rack is a click nobody asked for. Pausing does not.
- **METER** opens the bar: beats, unit, grouping, and how many steps that makes. It is the session's time signature — the metronome's — because there is one bar in this program.
- **Steps are counted from every downbeat of that bar.** A bar that is not a whole number of steps (7/8 at quarter steps is three and a half) ends on a short step, and steps past the sixteen the grid holds are rests, so step 1 is always the metronome's one.

9.3 Arranger

The arranger is a backing band from a chord chart: bass, drums, piano and guitar playing a progression in a style. A tab plays one artifact at a time, so switching the arranger on (its tab switch on the **ARR** tab of the generator box) switches that tab's arpeggiator and sequencer off. Switching it on does not start it:  does.



Figure 9.3 — The arranger on a SoundFont, playing the heterometric Tarkus chart: the band row with **SPLIT OUT**, the bar the band counts (5/4) and the chart, bar 5 lit.

-  plays the form from bar one,  leaves and comes back at the top of the bar it was left in,  stops, lets go of everything and rewinds (and takes the click, as in the sequencer).
-  comes in on the next downbeat of the metronome's bar, not on the press, so the band's one is the click's one — at most a bar's wait, half a bar on average. Pressed on the one (up to an eighth of a beat late) it starts that bar at once.
- **KEY** picks the tonic on a piano; the style button opens the list of every style (about two hundred, measured from real accompaniments), each described by how it plays — and **choosing one loads its time signature** (a waltz 3/4, a 6/8 blues 6/8) into **METER**, the metronome and the session, unless the chart names its own; **SEED** picks another interpretation of the same chart.
- **C–F–G / I–IV–V** switches the chart between chord letters and roman degrees — it rewrites the text.
- **BASS DRUMS PIANO GUITAR**: a click puts a musician in or out of the band, the wheel over the bar sets how loud. On a SoundFont every musician gets its own program (a kit for the drums, a bass for the bass); on other instruments the whole band plays the tab's one sound. The **right button** on a musician opens the SoundFont's programs for that musician alone (**BANK/PRESET**), with **AUTO** on top — the style's choice, named beside it; the pick is kept in the project. **n** in the **MIXER** on a tab playing a band levels it musician by musician, from the loudest each one played since the last time, and puts the tab back at 1.00: a font's kit can sit 15 dB over its piano, which the faders here — they set how hard each one plays — cannot make up. Loading an instrument only ever turns its tab down, never up: one middle C is not a band.
- **SPLIT OUT** (off by default) puts every musician on a strip of their own in the **MIXER** — after the tabs, before the groups — without opening any tab: a fader, a pan and a mute on that musician's channel of the tab's instrument, with its own meter. It needs the band to have channels to separate, which a SoundFont with the programs for it gives; reverb and chorus stay on the tab.
- **METER** is the bar the band counts; **SWING**, **RAND** and **PROB** vary the part the way the sequencer's knobs do.
- **LOAD** opens a .chord chart or a MIDI file (section 9.5); **EXPORT MIDI** writes the band out (section 9.5).
- The bars at the bottom of the box are the progression: a click opens the grid editor, a chord a subdivision, with **CHORD**, **GROUP**, **HOLD**, **BAR +**, **BAR -** and **BAND BAR**. Every bar is drawn once, however often the form plays it: form = turn turn turn is the twelve bars of the turn, and the light goes back to its first bar each time round, with the header saying which pass it is (**BAR 13 [TURN] 2/3**). Every chord of a bar is played by every musician: a chord the style's pattern has no hit for is struck where it starts.

9.4 Chord charts

A chart is a plain-text `.chord` file. The settings go on top, the bars below; `#` starts a comment. A fresh tab plays a twelve-bar blues in C — in the source tree it is `assets/default.chord`, a chart like any other.

```
key = D
style = strtrock
meter = 7/8 # or: time = 7/8
groups = 3+2+2 # or: grouping = 3 2 2
```

```
[A]
| Dm7 | G7:3 C:4 | Cmaj7 - | A7 |
[B]
| IIm7 | V7 | I | % |
form = A A B A
```

- `|` separates bars, and so does the end of a line. A bar with two chords splits it evenly; `C:3` gives a chord three of the bar's units.
- **A chart can change meter:** a bar that starts with a signature is in it, and so is every bar after it in the part until the next one — `| 5/4 Fm | Eb | 3/4 Db | 7/8 Cm |`. Each bar lasts what its signature says; a part starts back in the chart's bar. The source tree has an example, `assets/tarkus.chord`, for the `tarkus` style — which has a groove for each of the signatures it changes to.
- `-` holds the chord before it, `%` repeats the bar before.
- Letters (`C7`, `F#m7b5`, `Bbmaj9`) or degrees against the key (`I7`, `IVm7`, `VIIb7`).
- `[A]`, `[intro]`: parts; `form` says which are played in what order (without it, the order they were written in).
- `meter` and `groups`: the bar the chart is written in and how it is counted. They win over the session and the style, and **LOAD** hands them to the metronome and the grid. Groups that do not add up to the numerator are an error. A chart that changes meter counts each meter with a line of its own — `groups 7/8 = 2+2+3`, `groups 5/8 = 3+2` — and while the band plays, the click accents the grouping of the bar it is in; a meter with no line gets the downbeat alone.
- A chart that does not read leaves the band playing what it had, and says why on the box.

Styles. `style =` takes a style by its id, its name or the label the picker shows — `style = 42`, `style = drumnbass` and `style = Drum & Bass` are the same chart. The id is the number the **STYLE** list shows beside each style, and it does not change when styles are added. A name that matches none plays the first style. Every style is measured off a rhythm; the definition says what the band plays in it.

ID	style =	Name	Meter	Definition
1	16_beat	16 Beat	4/4	straight; bass on the changes; hi-hat in sixteenths
2	16bt_bld	16 Beat Ballad	4/4	straight; bass on the changes; hi-hat in sixteenths
3	16btshfl	16 Beat Shuffle	4/4	swing 67%; walking bass; hi-hat in eighths
4	50_srock	50s Rock	4/4	swing 42%; walking bass; hi-hat

ID	style =	Name	Meter	Definition
				in quarter-note triplets
5	50spnrck	50s Piano Rock	4/4	straight; walking bass; ride in eighths
6	60_s_8bt	60s 8 Beat	4/4	straight; bass on the changes; hi-hat in eighths
7	60_s_pop	60s Pop	4/4	straight; bass on the changes; hi-hat in eighths
8	60_srock	60s Rock	4/4	straight; bass on the changes; hi-hat in eighths
9	60_srock_2	60s Rock 2	4/4	straight; bass on the changes; hi-hat in eighths
10	60_sshfl	60s Shuffle	4/4	swing 67%; bass on the changes; hi-hat in eighths
11	60_ssoul	60s Soul	4/4	straight; walking bass; hi-hat in eighths
12	6_8_bld	6/8 Ballad	6/8	straight; bass on the changes; hi-hat in eighths
13	6_8_enka	6/8 Enka	4/4	swing 33%; walking bass; hi-hat in eighth-note triplets
14	6_8_pop	6/8 Pop	6/8	swing 8%; bass on the changes; hi-hat in eighths
15	6_8bld_1	6/8 Ballad 1	6/8	straight; walking bass; hi-hat in every 1.25 beats
16	6_8bld_2	6/8 Ballad 2	4/4	swing 33%; bass on the changes; hi-hat in eighth-note triplets
17	6_8bld_3	6/8 Ballad 3	6/8	straight; bass on the changes; hi-hat in eighths
18	6_8blues	6/8 Blues	6/8	straight; bass on the changes; hi-hat in eighths
19	6_8rkbld	6/8 Rock Ballad	6/8	straight; bass on the changes; hi-hat in eighths
20	90_s_bld	90s Ballad	4/4	straight; bass on the changes; hi-hat in eighths
21	a_gt_pop	Acoustic Guitar Pop	4/4	straight; bass on the changes; hi-hat in sixteenths
22	adani	Adani	4/4	straight; bass on the changes; hi-hat in eighths
23	amrcnrck	American Rock	4/4	straight; bass on the changes; hi-hat in eighths
24	argcmbia	Argentine Cumbia	4/4	swing 48%; bass on the changes; hi-hat in thirty-seconds
25	arpeggio1	Arpeggio 1	4/4	straight; bass on the changes; hi-hat in thirty-seconds
26	arpeggio2	Arpeggio 2	4/4	straight; bass on the changes; hi-hat in thirty-seconds
27	arpeggio3	Arpeggio 3	4/4	straight; bass on the changes; hi-hat in thirty-seconds
28	bachata	Bachata	4/4	straight; bass on the changes; hi-hat in thirty-seconds

ID	style =	Name	Meter	Definition
29	baladi	Baladi	4/4	swing 50%; bass on the changes; hi-hat in eighths
30	banda	Banda	4/4	straight; bass on the changes; hi-hat in thirty-seconds
31	beguine	Beguine	4/4	swing 38%; bass on the changes; hi-hat in thirty-seconds
32	beijing	Beijing	2/4	straight; bass on the changes; ride in quarters
33	bhajan	Bhajan	4/4	swing 21%; walking bass; hi-hat in thirty-seconds
34	bhangra	Bhangra	4/4	swing 65%; bass on the changes; ride in quarters
35	bluegras	Bluegrass	4/4	swing 67%; bass on the changes; hi-hat in thirty-seconds
36	bluesbld	Blues Ballad	6/4	straight; bass on the changes; hi-hat in quarters
37	bolero	Bolero	4/4	swing 48%; bass on the changes; hi-hat in half notes
38	boogie	Boogie	4/4	swing 33%; walking bass; hi-hat in thirty-seconds
39	bosanova	Bossa Nova	4/4	straight; walking bass; hi-hat in half notes
40	brushbld	Brush Ballad	4/4	straight; bass on the changes; hi-hat in thirty-seconds
41	calypso	Calypso	4/4	straight; bass on the changes; hi-hat in thirty-seconds
42	caucasia	Caucasia	4/4	swing 58%; bass on the changes; hi-hat in eighths
43	cha_cha	Cha Cha	4/4	swing 48%; bass on the changes; hi-hat in thirty-seconds
44	chinspop	Chinese Pop	4/4	swing 62%; bass on the changes; hi-hat in eighth-note triplets
45	cnty_8bt	Country 8 Beat	4/4	straight; bass on the changes; hi-hat in eighths
46	cnty_bld	Country Ballad	4/4	straight; walking bass; hi-hat in eighths
47	cnty_pop	Country Pop	4/4	straight; bass on the changes; hi-hat in eighths
48	cntyshfl	Country Shuffle	4/4	swing 29%; bass on the changes; hi-hat in every 0.625 beats
49	cntywltz	Country Waltz	3/4	straight; bass on the changes; hi-hat in quarters
50	cumbia_1	Cumbia 1	4/4	swing 50%; walking bass; hi-hat in sixteenths
51	cumbia_2	Cumbia 2	4/4	swing 50%; bass on the changes; hi-hat in eighth-note triplets
52	dadra	Dadra	6/4	swing 50%; walking bass; hi-hat in sixteenths
53	daizu	Daizu	3/4	straight; bass on the changes; hi-

ID	style =	Name	Meter	Definition
				hat in thirty-seconds
54	dandiya	Dandiya	4/4	swing 31%; walking bass; hi-hat in sixteenths
55	dangdut	Dangdut	4/4	swing 50%; bass on the changes; hi-hat in quarters
56	discopop	Disco Pop	4/4	swing 58%; bass on the changes; hi-hat in eighths
57	dixie	Dixie	4/4	swing 33%; walking bass; hi-hat in quarter-note triplets
58	dongbeiy	Dongbeiy	4/4	straight; walking bass; hi-hat in thirty-seconds
59	drngense	Drngense	4/4	straight; walking bass; hi-hat in quarters
60	drumbass	Drum & Bass	4/4	straight; bass on the changes; hi-hat in sixteenths
61	dscosoul	Disco Soul	4/4	swing 62%; walking bass; hi-hat in eighths
62	e_hiphop	Electro Hip Hop	4/4	straight; bass on the changes; hi-hat in eighths
63	e_hiphop_2	Electro Hip Hop 2	4/4	straight; bass on the changes; hi-hat in eighths
64	easy_bld	Easy Ballad	4/4	straight; bass on the changes; hi-hat in sixteenths
65	elec_bld	Electric Ballad	4/4	straight; bass on the changes; hi-hat in eighths
66	enka	Enka	4/4	straight; bass on the changes; hi-hat in eighths
67	ep_bld_1	Electric Piano Ballad 1	4/4	straight; bass on the changes; hi-hat in sixteenths
68	ep_bld_2	Electric Piano Ballad 2	4/4	straight; bass on the changes; hi-hat in sixteenths
69	euro_pop	Euro Pop	4/4	straight; bass on the changes; hi-hat in eighths
70	f_gospel	Fast Gospel	4/4	straight; walking bass; hi-hat in eighths
71	fastsoul	Fast Soul	4/4	straight; bass on the changes; hi-hat in quarters
72	fingcnty	Fingerpicking Country	4/4	straight; bass on the changes; hi-hat in eighths
73	folklore	Folklore	4/4	straight; bass on the changes; hi-hat in thirty-seconds
74	forro	Forro	4/4	straight; bass on the changes; hi-hat in thirty-seconds
75	fox_trot	Foxtrot	4/4	swing 33%; bass on the changes; hi-hat in quarter-note triplets
76	fr_waltz	French Waltz	3/4	swing 33%; bass on the changes; hi-hat in quarters
77	fstbband	Fast Big Band	4/4	swing 42%; walking bass; hi-hat in quarter-note triplets

ID	style =	Name	Meter	Definition
78	funk16bt	Funk 16 Beat	4/4	swing 50%; bass on the changes; hi-hat in eighths
79	funk_8bt	Funk 8 Beat	4/4	swing 50%; bass on the changes; hi-hat in eighths
80	funkshfl	Funk Shuffle	4/4	swing 67%; bass on the changes; hi-hat in eighth-note triplets
81	garba	Garba	4/4	swing 33%; walking bass; ride in quarters
82	grmmrch	German March	4/4	swing 50%; walking bass; ride in sixteenths
83	gtr_8bt	Guitar 8 Beat	4/4	straight; bass on the changes; hi-hat in eighths
84	guangdon	Guangdong	4/4	straight; bass on the changes; hi-hat in thirty-seconds
85	hawaiian	Hawaiian	4/4	swing 33%; bass on the changes; hi-hat in half notes
86	hip_hop1	Hip Hop 1	4/4	swing 50%; bass on the changes; hi-hat in eighths
87	hip_hop1_2	Hip Hop 1 2	4/4	swing 50%; bass on the changes; hi-hat in eighths
88	hip_hop2	Hip Hop 2	4/4	straight; walking bass; hi-hat in sixteenths
89	hip_hop2_2	Hip Hop 2 2	4/4	straight; walking bass; hi-hat in sixteenths
90	huangmei	Huangmei	4/4	straight; bass on the changes; hi-hat in eighths
91	indipop1	Indian Pop 1	4/4	swing 12%; bass on the changes; hi-hat in sixteenths
92	indipop2	Indian Pop 2	4/4	swing 23%; walking bass; hi-hat in every 0.5833 beats
93	jhaptaal	Jhaptaal	10/4	swing 58%; bass on the changes; hi-hat in thirty-seconds
94	jiangnan	Jiangnan	4/4	swing 50%; walking bass; hi-hat in sixteenths
95	jingju	Jingju	4/4	straight; bass on the changes; hi-hat in thirty-seconds
96	jz_waltz	Jazz Waltz	3/4	swing 33%; bass on the changes; hi-hat in quarters
97	jzcombo1	Jazz Combo 1	4/4	swing 33%; walking bass; ride in quarter-note triplets
98	jzcombo2	Jazz Combo 2	4/4	swing 33%; bass on the changes; hi-hat in half notes
99	jzcombo3	Jazz Combo 3	4/4	swing 42%; bass on the changes; hi-hat in half notes
100	keharwa	Keharwa	4/4	straight; bass on the changes; hi-hat in thirty-seconds
101	khaliji	Khaliji	4/4	straight; bass on the changes; hi-hat in sixteenths
102	kroncong	Kroncong	4/4	straight; walking bass; ride in half

ID	style =	Name	Meter	Definition
				notes
103	kwaito	Kwaito	4/4	swing 12%; walking bass; hi-hat in dotted eighths
104	kwaito_2	Kwaito 2	4/4	swing 12%; walking bass; hi-hat in dotted eighths
105	ltinfusn	Latin Fusion	4/4	straight; walking bass; ride in eighths
106	ltinrock	Latin Rock	4/4	swing 42%; bass on the changes; hi-hat in eighths
107	malfouf	Malfouf	4/4	swing 67%; bass on the changes; hi-hat in thirty-seconds
108	mambo	Mambo	4/4	swing 50%; bass on the changes; hi-hat in thirty-seconds
109	march	March	4/4	swing 33%; walking bass; hi-hat in quarters
110	mdrn_bld	Modern Ballad	4/4	straight; walking bass; hi-hat in sixteenths
111	mdrn_r_b	Modern R&B	4/4	swing 65%; bass on the changes; hi-hat in eighths
112	mdrncnty	Modern Country	4/4	swing 67%; bass on the changes; hi-hat in eighths
113	melow8bt	Mellow 8 Beat	4/4	straight; bass on the changes; hi-hat in eighths
114	menggu	Menggu	4/4	swing 58%; walking bass; hi-hat in thirty-seconds
115	merengue	Merengue	4/4	swing 50%; bass on the changes; hi-hat in sixteenths
116	miaoazu	Miaoazu	4/4	straight; bass on the changes; hi-hat in thirty-seconds
117	midbband	Mid Big Band	4/4	straight; bass on the changes; ride in quarters
118	mus	Mus	5/8	straight; walking bass; hi-hat in sixteenths
119	n_o_r_r	New Orleans Rock & Roll	4/4	swing 8%; walking bass; hi-hat in half notes
120	norteno	Norteno	4/4	straight; bass on the changes; hi-hat in quarters
121	ochswing	Orchestral Swing	4/4	swing 35%; walking bass; hi-hat in half notes
122	old_rock	Old Rock	4/4	straight; bass on the changes; hi-hat in eighths
123	old_soul	Old Soul	4/4	straight; bass on the changes; hi-hat in eighths
124	oldie8bt	Oldies 8 Beat	4/4	swing 33%; bass on the changes; hi-hat in quarter-note triplets
125	orgnrock	Organ Rock	4/4	swing 50%; bass on the changes; hi-hat in eighths
126	pagode	Pagode	4/4	swing 8%; bass on the changes; hi-hat in thirty-seconds

ID	style =	Name	Meter	Definition
127	pasillo	Pasillo	3/4	straight; bass on the changes; hi-hat in thirty-seconds
128	pasodble	Paso Doble	4/4	straight; walking bass; ride in quarters
129	pianor_r	Piano Rock & Roll	4/4	straight; bass on the changes; hi-hat in thirty-seconds
130	pno8beat	Piano 8 Beat	4/4	swing 50%; bass on the changes; hi-hat in sixteenths
131	pnobld_1	Piano Ballad 1	4/4	straight; bass on the changes; hi-hat in eighths
132	pnobld_2	Piano Ballad 2	4/4	straight; walking bass; hi-hat in eighths
133	pnomrch1	Piano March 1	4/4	straight; walking bass; hi-hat in thirty-seconds
134	pnomrch2	Piano March 2	4/4	straight; bass on the changes; hi-hat in thirty-seconds
135	pnowaltz	Piano Waltz	3/4	straight; bass on the changes; hi-hat in quarters
136	pnrckbld	Piano Rock Ballad	4/4	straight; bass on the changes; ride in eighths
137	polka	Polka	4/4	straight; walking bass; hi-hat in quarters
138	polwaltz	Polka Waltz	3/4	swing 40%; bass on the changes; hi-hat in quarters
139	pop	Pop	4/4	straight; bass on the changes; hi-hat in eighths
140	pop_bld	Pop Ballad	4/4	swing 50%; bass on the changes; hi-hat in eighths
141	pop_rock	Pop Rock	4/4	swing 50%; bass on the changes; hi-hat in eighths
142	pop_shfl	Pop Shuffle	4/4	swing 33%; bass on the changes; hi-hat in eighth-note triplets
143	popregae	Pop Reggae	4/4	swing 65%; bass on the changes; hi-hat in eighth-note triplets
144	punta	Punta	4/4	swing 50%; walking bass; hi-hat in thirty-seconds
145	qinqiang	Qinqiang	4/4	straight; walking bass; hi-hat in thirty-seconds
146	quickstp	Quickstep	4/4	swing 33%; bass on the changes; hi-hat in quarter-note triplets
147	r_b	R&B	4/4	swing 50%; bass on the changes; hi-hat in eighths
148	r_b_bld	R&B Ballad	4/4	swing 50%; bass on the changes; hi-hat in quarters
149	r_r	Rock & Roll	4/4	swing 33%; walking bass; hi-hat in half notes
150	ragtime	Ragtime	4/4	straight; bass on the changes; hi-hat in thirty-seconds
151	rckblues	Rock Blues	4/4	straight; bass on the changes; hi-

ID	style =	Name	Meter	Definition
				hat in eighths
152	regeton1	Reggaeton 1	4/4	swing 50%; bass on the changes; hi-hat in thirty-seconds
153	regeton2	Reggaeton 2	4/4	straight; bass on the changes; hi-hat in sixteenths
154	reggae	Reggae	4/4	swing 62%; walking bass; hi-hat in eighths
155	rhumba	Rhumba	4/4	swing 50%; walking bass; ride in eighths
156	rock	Rock	4/4	swing 8%; bass on the changes; hi-hat in quarters
157	rock_bld	Rock Ballad	4/4	swing 50%; walking bass; hi-hat in quarters
158	rockwltz	Rock Waltz	6/4	swing 17%; bass on the changes; hi-hat in quarters
159	ruchnsn1	Ruchnsn1	4/4	straight; walking bass; hi-hat in quarters
160	ruchnsn2	Ruchnsn2	4/4	straight; walking bass; hi-hat in eighths
161	rupak	Rupak	7/4	swing 50%; walking bass; hi-hat in sixteenths
162	s_gospel	Slow Gospel	6/4	straight; bass on the changes; hi-hat in quarters
163	salsa_1	Salsa 1	4/4	straight; walking bass; hi-hat in thirty-seconds
164	salsa_2	Salsa 2	4/4	straight; walking bass; hi-hat in thirty-seconds
165	samba_1	Samba 1	4/4	straight; bass on the changes; ride in sixteenths
166	samba_2	Samba 2	4/4	swing 8%; bass on the changes; hi-hat in quarters
167	schlager	Schlager	4/4	straight; bass on the changes; hi-hat in eighths
168	scrswing	Scrswing	4/4	straight; bass on the changes; hi-hat in half notes
169	serenade	Serenade	4/4	straight; walking bass; hi-hat in thirty-seconds
170	shfblues	Shuffle Blues	4/4	swing 35%; walking bass; hi-hat in quarters
171	shflrock	Shuffle Rock	4/4	swing 33%; bass on the changes; hi-hat in quarter-note triplets
172	shouka_1	Shouka 1	4/4	straight; bass on the changes; hi-hat in quarters
173	shouka_2	Shouka 2	3/4	straight; bass on the changes; hi-hat in thirty-seconds
174	sirtaki	Sirtaki	4/4	straight; walking bass; ride in quarters
175	ska	Ska	4/4	swing 54%; walking bass; hi-hat in quarters

ID	style =	Name	Meter	Definition
176	slow_bld	Slow Ballad	4/4	straight; bass on the changes; hi-hat in eighths
177	slowbld1	Slow Ballad 1	4/4	straight; bass on the changes; hi-hat in eighths
178	slowbld2	Slow Ballad 2	4/4	straight; walking bass; hi-hat in eighths
179	slowrock	Slow Rock	4/4	swing 33%; bass on the changes; hi-hat in half notes
180	slowsoul	Slow Soul	6/8	swing 67%; bass on the changes; hi-hat in eighths
181	slwbband	Slow Big Band	4/4	straight; walking bass; hi-hat in half notes
182	slwblues	Slow Blues	4/4	swing 33%; bass on the changes; hi-hat in eighth-note triplets
183	slwbossa	Slow Bossa	4/4	straight; bass on the changes; hi-hat in half notes
184	slwswing	Slow Swing	4/4	straight; walking bass; hi-hat in half notes
185	slwwaltz	Slow Waltz	3/4	straight; bass on the changes; hi-hat in quarters
186	strd_pno	Stride Piano	4/4	straight; bass on the changes; hi-hat in thirty-seconds
187	strqrtet	String Quartet	4/4	straight; bass on the changes; hi-hat in thirty-seconds
188	strt_8bt	Straight 8 Beat	4/4	straight; bass on the changes; hi-hat in eighths
189	strtrock	Straight Rock	4/4	straight; bass on the changes; hi-hat in eighths
190	swing	Swing	4/4	swing 33%; walking bass; ride in quarter-note triplets
191	synthpop	Synth Pop	4/4	straight; bass on the changes; hi-hat in eighths
192	tango	Tango	4/4	swing 46%; bass on the changes; hi-hat in thirty-seconds
193	tarkus	Tarkus (heterometric)	5/4	straight; bass on the changes; ride in eighths
194	teccmbia	Techno Cumbia	4/4	swing 52%; walking bass; hi-hat in thirty-seconds
195	tech_pop	Techno Pop	4/4	straight; bass on the changes; hi-hat in eighths
196	teentaal	Teentaal	4/4	swing 50%; walking bass; ride in eighths
197	tex_mex	Tex Mex	4/4	swing 54%; bass on the changes; hi-hat in quarters
198	trance	Trance	4/4	straight; bass on the changes; hi-hat in sixteenths
199	uk_beat	UK Beat	4/4	straight; bass on the changes; hi-hat in eighths
200	unplgbld	Unplugged Ballad	4/4	straight; bass on the changes; hi-

ID	style =	Name	Meter	Definition
				hat in thirty-seconds
201	valenato	Vallenato	4/4	straight; bass on the changes; hi-hat in sixteenths
202	vienwltz	Viennese Waltz	3/4	swing 50%; walking bass; hi-hat in thirty-seconds
203	waltz_1	Waltz 1	3/4	straight; bass on the changes; hi-hat in quarters
204	waltz_2	Waltz 2	3/4	straight; bass on the changes; hi-hat in thirty-seconds
205	waltz_3	Waltz 3	3/4	straight; bass on the changes; hi-hat in thirty-seconds
206	waltz_4	Waltz 4	3/4	straight; bass on the changes; hi-hat in thirty-seconds
207	xinjiang	Xinjiang	4/4	swing 46%; bass on the changes; hi-hat in sixteenths
208	xmassong	Christmas Song	4/4	straight; bass on the changes; hi-hat in half notes
209	xmaswltz	Christmas Waltz	3/4	straight; bass on the changes; hi-hat in quarters
210	yaozu	Yaozu	4/4	straight; bass on the changes; hi-hat in thirty-seconds
211	yuju	Yuju	4/4	swing 31%; bass on the changes; hi-hat in thirty-seconds
212	zangzu	Zangzu	4/4	straight; bass on the changes; hi-hat in thirty-seconds

9.5 MIDI import and export

EXPORT MIDI asks for a folder and then a file name (.mid is added; an existing file is asked about). The file is type 1: a conductor track with the tempo, the time signature, a marker at every chord change and the chart itself, then one track per musician on the channel a GM player expects — drums on 10 — with the style's program.

Levels. Inside a tab the whole band plays one instrument, so the parts are baked to sit under each other. In the file each musician has a channel of its own, so each part is brought to its own average velocity — drums 96, bass 92, piano 80, guitar 74, accents kept — and each musician's fader goes on the channel volume (CC 7; 100 at full), where the DAW's mixer can find it.

LOAD opens .chord, .mid and .midi. A MIDI file choz wrote opens as exactly the chart that wrote it. Any other is listened to: one chord a bar, or two when the halves of the bar are plainly different, with the root taken from the bass register and the drum channel ignored. It is a heuristic — a melody full of passing notes can split a bar that is one chord — and the grid editor is where to correct it.

Note. The arranger reads and writes **two to four instruments** (channels with notes, drums included). A single melody or a full GM song is not a backing and is refused; so is exporting a band of one — put a second musician in first.

9.6 The arranger and the click

While an arranger plays, the metronome **follows its bar**: the chart's meter, else the style's own (a waltz clicks in three),

with its grouping. The sequencer's grid follows too, and the session's own time signature comes back when the band stops. In a chart that changes meter the click changes with every bar, and counts its one from that bar's downbeat. **FOLLOW ARR** in the metronome's menu turns this off; setting BEATS or UNIT by hand turns it off as well, because that is the player taking the count back.

10. Transport, tempo and metronome

10.1 Transport

F7 starts and stops the rack, **F8** arms automation recording, **F9** fires panic — the same three controls that sit on the menu bar. The transport is what every sequencer, every synced delay and the automation count against.

10.2 Metronome

F6 switches the click on; the ▼ beside **MET** opens its options: click, tempo, beats and unit of the bar, grouping, sound, output and level. It keeps counting with the transport stopped — which is usually when it is needed — and it counts the same tempo the tempo-synced plugins read.

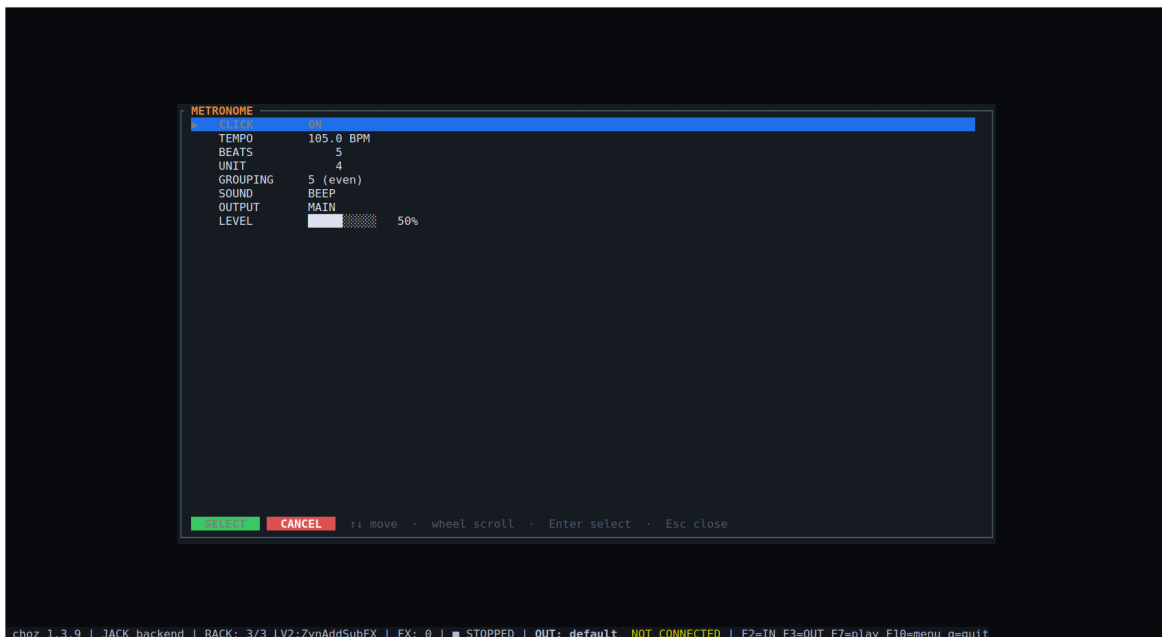


Figure 10.1 — The metronome's options: click, tempo, time signature, grouping, sound, output and level.

Its last row, **FOLLOW ARR**, makes the click count the bar of a playing arranger (section 9.6). Switching the click on never restarts the count: it joins on the beat the music is already on, and a sequencer's or an arranger's ■ switches it off with the band.

10.3 TAP

TAP sits immediately left of **MET**: four clicks set the transport's tempo. It is the same count the arpeggiator's own TAP does — the average of the last four intervals, with a gap longer than two seconds starting a new count, because that was somebody stopping rather than playing very slowly.

- The button **lights on every tap**: playing a tempo in is a gesture with no other feedback, and a button that never moves cannot say whether the click landed.
- On the **fourth tap the metronome starts by itself** — counting four in is counting for a click, and the click you were counting for is the one you want to hear.
- Taps are quarter notes whatever division the arpeggiator is running, and the tempo they set is the transport's, so everything synced follows.

Tip. TAP is also a MIDI learn target, so a pad on a controller can play the tempo in without reaching for the mouse. TAP has a colour of its own — a muted teal, softer than the metronome's gold — so it reads as a button to press rather than as part of the MET switch.

11. Mixer, routing and views

11.1 The mixer

The **MIXER** panel shows the whole rack at once as channel strips: one vertical fader per output channel with a link between the two, pan, and the **O M S** row under the fader (output, mute, solo).

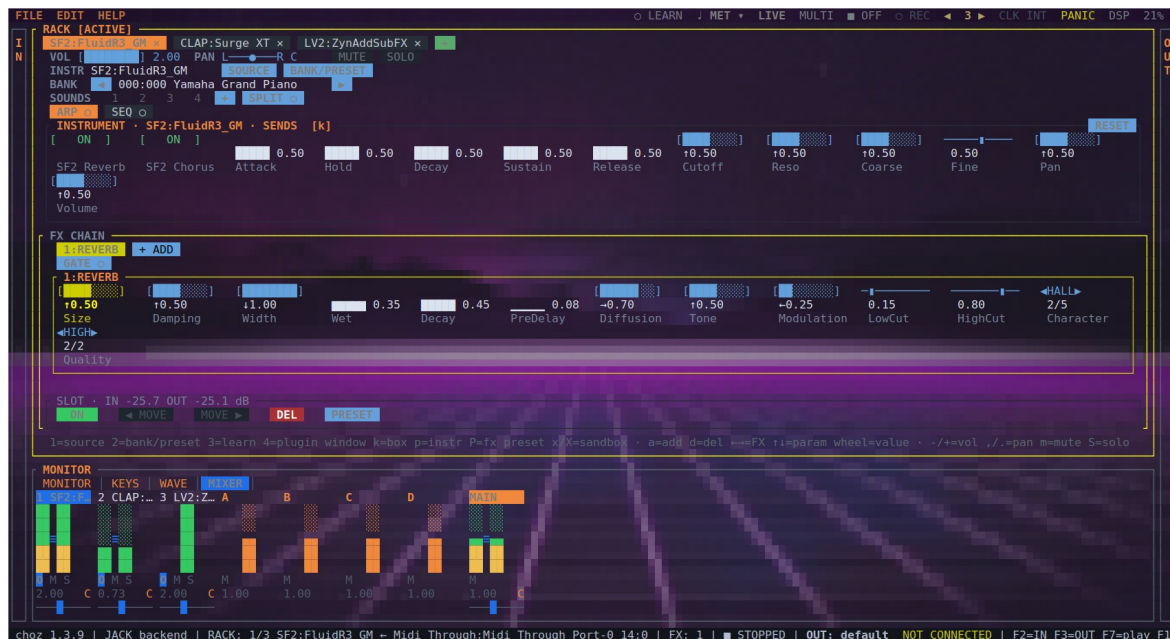


Figure 11.1 — MIXER: tabs, groups and the main mix, each strip with its faders, pan and O M S.

Key	Action
Tab	Focus the mixer while it is on screen (clicking a strip does the same)
↑ ↓ / wheel	The strip's level, in the same step VOL uses in the rack
← →	Next strip: the tabs, then the four groups, then the main mix
l / k	Link or unlink the strip's two channels / choose which side the arrows move
, . / m / S	Pan / mute / solo

A split-out band. With the arranger's **SPLIT OUT** up (section 9.3), each musician of a SoundFont tab's band gets a strip of its own after the tabs and before the groups — 1· BASS, 1· DRUMS — with one fader, a pan, a mute and its own meter, acting on that musician's channel inside the tab before the tab's fader. No tab is added. The mouse and the wheel move them; the arrows walk the tabs and the desk and skip them, and they are not MIDI-learn targets yet.

11.2 Groups and the main mix

Four groups (A, B, C, D) sit between the tabs and the main mix, for a submix with one fader on it. The main is a stereo pair with its two sides tied by default and a balance rather than a pan: a centred main is untouched, and the sides are broken apart only to trim one wedge against the other.

11.3 Direct outs

The **O** button on a strip opens the list of where that tab sums: **MAIN**, the four groups, and **DIRECT n** — a pair of JACK ports of the tab's own with nothing on the other end, for a DAW to patch into. The main fader only ever touches the first pair, so a tab on its direct out is off the master mix by construction.

- **One direct per tab, at a fixed place.** Tab 1 owns the first pair, tab 2 the second, and each strip offers only its own — a shared list let two tabs pick the same one and land on the same ports. The place never moves: a layout that shifted when the rack changed shape would break the patch in the DAW that the feature exists for.
- **As wide as the tab is.** A mono tab routes to a single port — what a DAW wants to record off a guitar is one track, not two identical ones. It becomes stereo the moment something in the chain has a reason for the sides to differ: a reverb tail, a chorus spread, a pan, a ping-pong delay, or any hosted plugin. A bypassed effect does not count, and the width is re-measured when the chain or the jack changes.
- The strip's destination cell says where it is going: D1...D8 for a direct out, the group's letter for a group, nothing for the main mix.

11.4 The views

View	What it answers
MONITOR	Did the note arrive, what came out, how loud: the MIDI log, the spectrum and the meters, side by side.
KEYS	What is sounding right now — held notes, bend and modulation, one colour per tab.
WAVE	The waveform of what is playing.
MIXER	The whole desk: every tab, the four groups and the main mix.
HARMONICS	Not a panel tab but a view of its own (h): the harmonic series of an instrument that publishes it, edited in place.

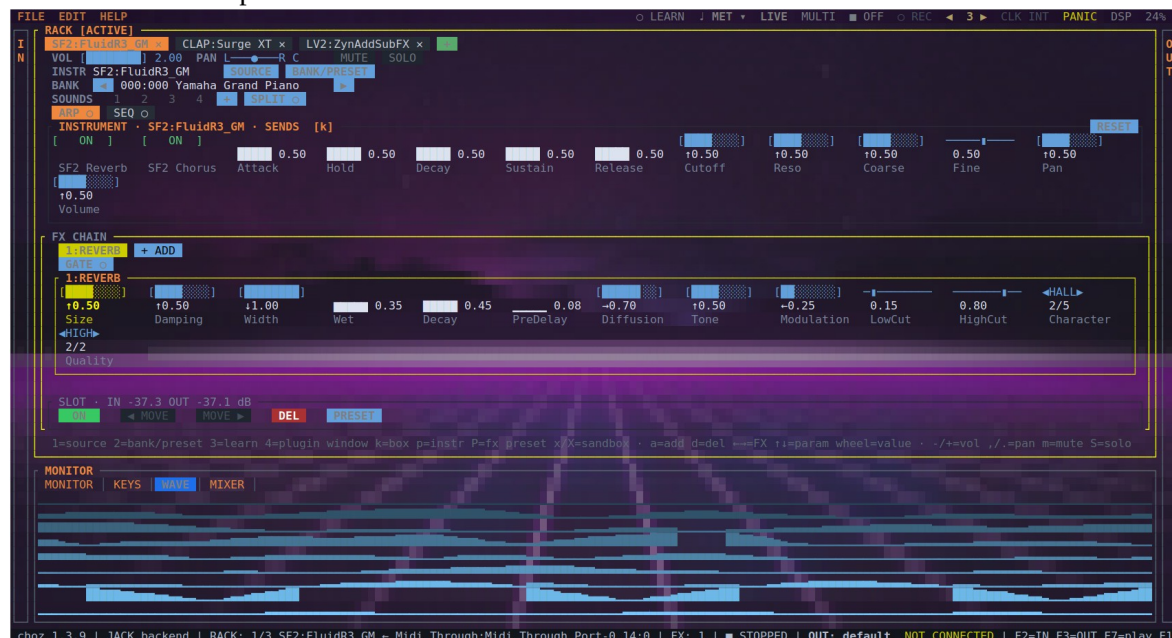


Figure 11.2 — The WAVE panel with a SoundFont's signal going through the internal reverb.

12. Projects

F10 opens the menu bar. **FILE** makes, saves and opens projects; **EDIT** leads to Settings and to rescanning the plugin paths; **HELP** shows the About box.

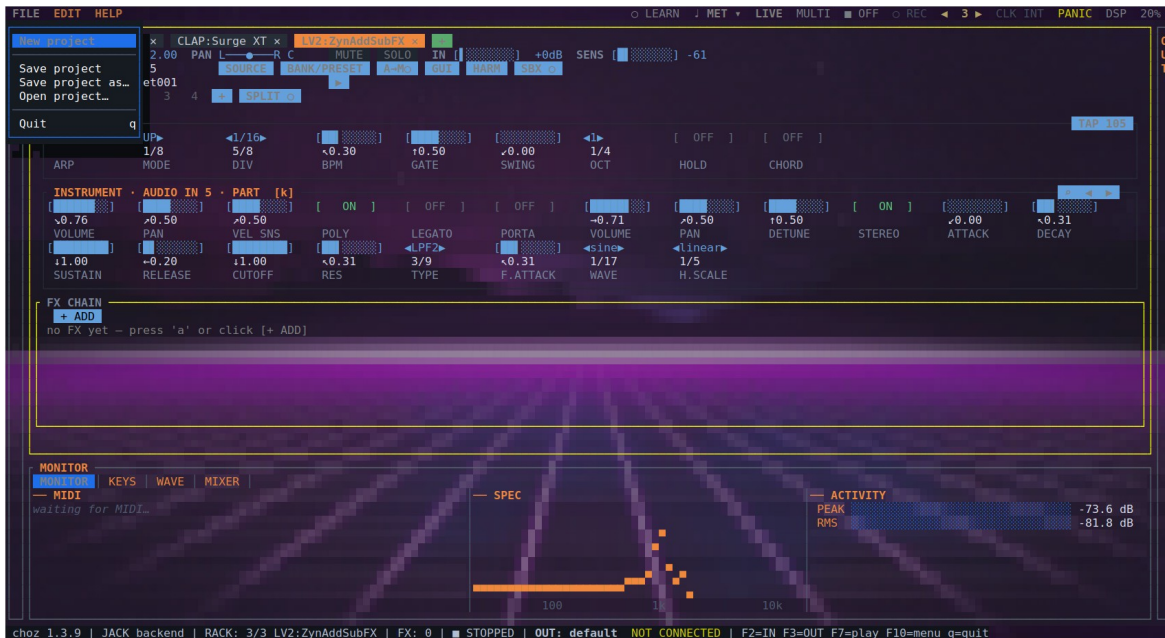


Figure 12.1 — The FILE menu: new project, save, save as, open and quit.

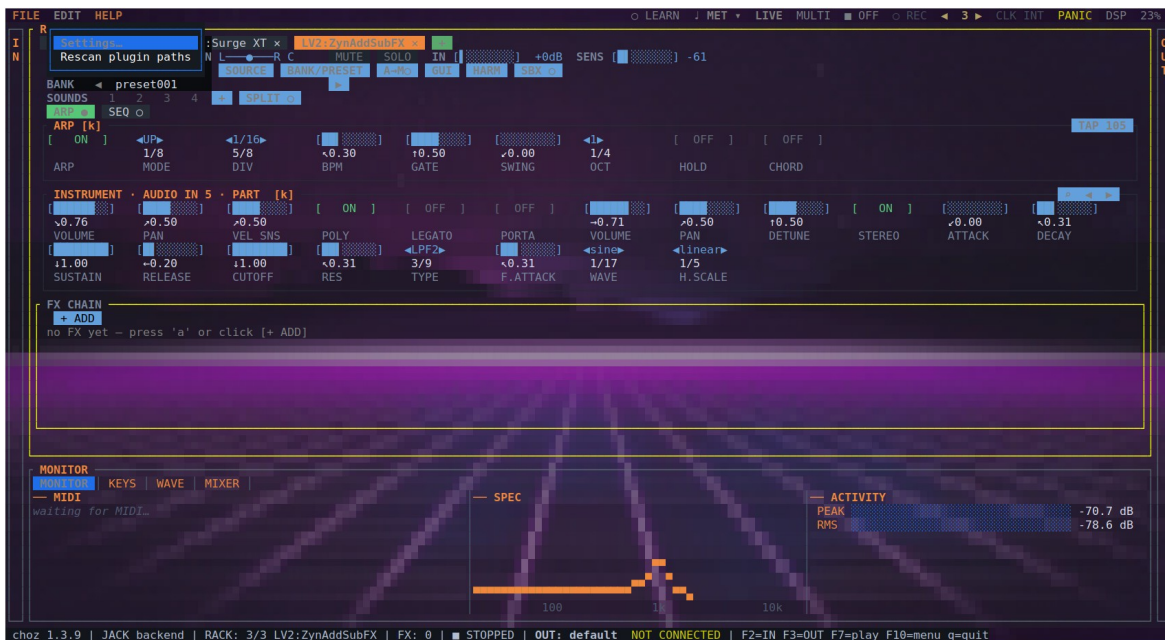


Figure 12.2 — The EDIT menu: Settings and rescanning the plugin paths.

A project is readable YAML that can be kept under version control. Besides the rack and its routing it carries each plugin's own state (section 7.5). Looper takes are saved as WAV inside <project>. loops/ next to the file: moving a project means moving the .yaml and its folder. A take recorded at another sample rate is resampled on open, keeping its pitch and its length in seconds.

13. Preferences

EDIT → Settings has three tabs: **AUDIO**, **THEME** and **LANGUAGE**.

13.1 Audio

The engine: backend (AUTO, ALSA or JACK), device, input, sample rate, buffer size, tempo, time signature, feedback guard, polyphony, SF2 engine, the resulting latency and the PipeWire quantum.

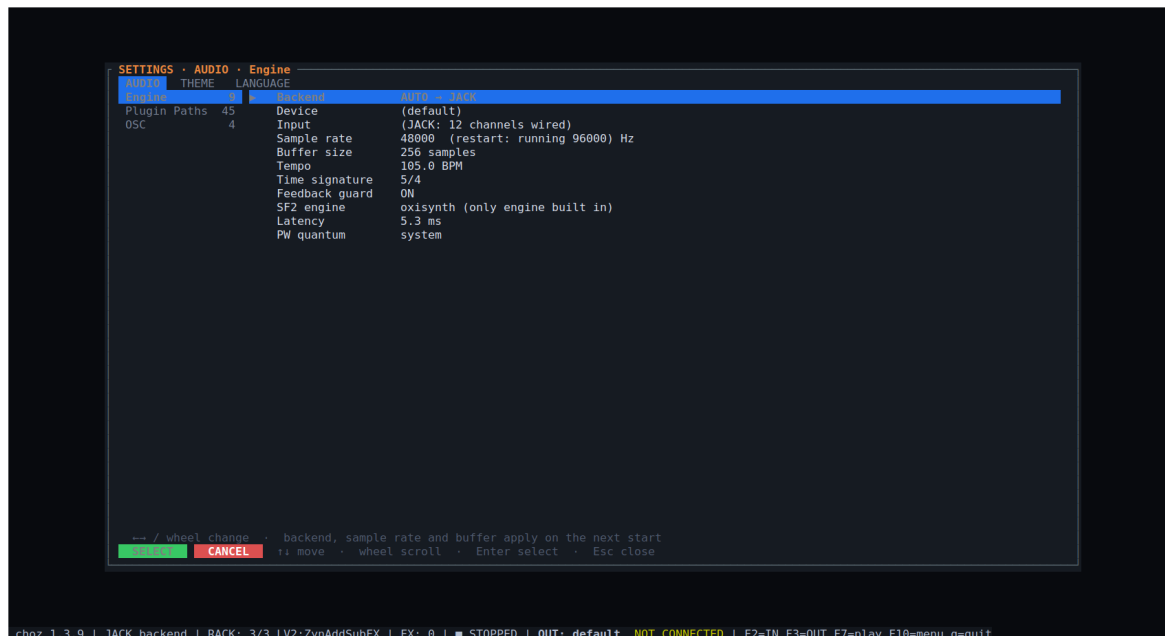


Figure 13.1 — Settings · AUDIO: backend, device, rate, buffer and the latency they add up to.

Note. The backend, the sample rate and the buffer size take effect on the next start. Choosing **JACK** is what puts `choz:out_*`, `choz:in_*`, `choz:midi_in` and `choz:midi_out` on the graph.

Polyphony is how many voices one instrument may sound at once — a SoundFont tab and a sampler tab each: 16 to 1024, 256 by default (FluidSynth's own). It is a CPU budget rather than a musical limit: a voice costs only while it sounds, but a held sustain pedal on a layered preset fills the pool and keeps it full. It applies to instruments loaded after the change. On a 96 kHz graph with several SoundFont tabs under the pedal, 128 is the safer choice.

13.2 Plugin paths

Per format, every search folder with how many plugins were found in it, or (missing). The buttons edit, add, browse, remove or restore the defaults. The format's environment variable, when set, replaces the list (section 15).

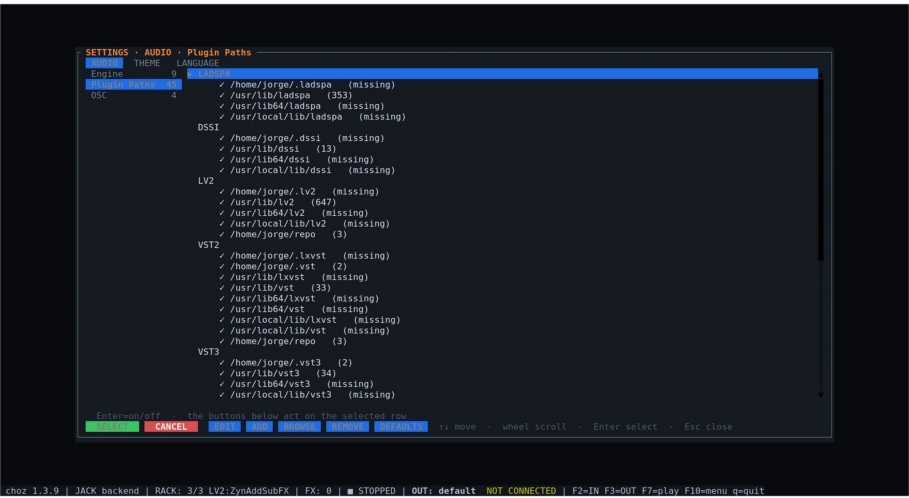


Figure 13.2 — The search paths per format, with how many plugins each holds.

13.3 Theme and wallpaper

Every change here applies live and is saved.

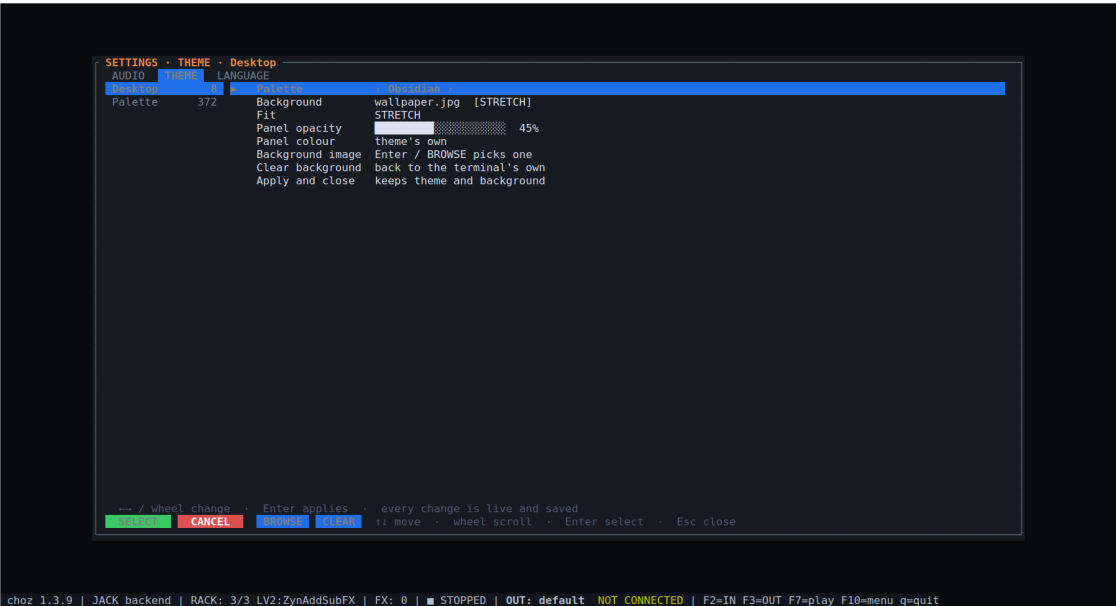


Figure 13.3 — Settings · THEME: palette, wallpaper, fit and panel opacity.

Row	What it does
Palette	372 colour schemes — choz's own eleven first, then the Gogh terminal schemes. ← → try one in place without leaving the row; Enter opens the full list at the scheme currently worn.
Background	Cycles the terminal's own background and the scheme's flat colour.
Pick image	Opens the file browser on the project's assets/ to set a wallpaper; the installed wallpapers live there.
Fit	Stretch or tile. Present only while an image is set.
Panel opacity	How strongly the panel colour washes over the wallpaper. Absent on the terminal's own background: choz cannot read that colour, so there is nothing to be translucent against.

Panel colour Which colour that wash is — the scheme's own, or any palette entry.

Clear
background Back to the terminal's own background.

On kitty, Ghostty and WezTerm the wallpaper is drawn with the terminal's graphics protocol; elsewhere it is painted with cell colours, and `CHOZ_KITTY_BG=0` forces that path. The picture is re-sent only once the window size has settled, so dragging a window edge does not re-scale a large image on every frame.

13.4 Language

The **LANGUAGE** tab switches the interface language live, with no restart. Nine are available:

Language Code

English	en
Español	es
Português	pt
Français	fr
Italiano	it
Deutsch	de
Русский	ru
日本語	ja
中文	zh

At first start choz takes the language from the environment (`LC_ALL`, then `LC_MESSAGES`, then `LANG`) and falls back to English. What you pick here is saved in `ui.json` and wins from then on.

14. Reference: keyboard shortcuts

Key	Where	Action
F2 / F3	anywhere	The IN and OUT drawers
Enter / Space	a drawer row	Bind that input to the active tab (or unbind it)
c / r	IN drawer	Connect or disconnect the port / rescan
F4	anywhere	LIVE ↔ MULTI
F5	anywhere	Bottom panel: MONITOR / KEYS / WAVE / MIXER
F6	anywhere	Metronome (the ▼ opens its options)
F7 / F8 / F9	anywhere	Start the rack / arm automation / panic
F10	anywhere	The menu bar
[/]	rack	Previous / next tab
1 or i	rack	Change the tab's instrument
2 or b	rack	Bank and preset
p	rack	The instrument's parameter list
/	rack (instrument)	Search a knob by name
PgUp / PgDn	rack	Page the knob box (learned CCs move with it)
h	rack	HARMONICS, for an instrument that publishes them
k	rack	Move the cursor between instrument and effects
a / d	rack	Add an effect / delete the selected one
Space	rack (FX)	Switch the effect on or off
g / G	rack	The instrument's own window / the selected effect's
x / X	rack	Run the instrument / the effect isolated
l	rack	MIDI learn
3	rack	MIDI learn, picking the control first
A	anywhere	The active tab's arpeggiator
C / c	rack (FX)	The chord's keyboard / the gate from another tab
v	rack	Split: which sound each octave plays
< > and ; :	rack	Input trim and A → M sensitivity on an audio tab
- / +	rack	The tab's volume
, / .	rack, MIXER	Pan
m / S	rack, MIXER	Mute / solo
n / N	rack, MIXER	Level the tab / the whole rack
↑ ↓ / wheel	MIXER	The strip's level
← →	MIXER	Next strip: tabs, groups, main
Tab	MIXER	Focus the mixer
P	anywhere	Panic: silence every note
q	anywhere	Quit

TAP, LEARN, the **O** button on a strip and the metronome's ▼ are mouse controls.

15. Reference: environment variables and files

Variable	Effect
PIPEWIRE_LATENCY / PIPEWIRE_QUANTUM	Set by choz from the configured buffer before it opens the JACK client
CHOZ_SANDBOX_GUI=1	Isolate every plugin with a window, not only the ones that have crashed before
CHOZ_PROBE_RUNS=N	How many times a plugin is probed before it is called safe (3 by default)
CHOZ_KITTY_BG=0	Draw the wallpaper with cell colours instead of the terminal's graphics protocol
CHOZ_CLAP_STRICT_TEARDOWN / CHOZ_LV2_STRICT_TEARDOWN	Destroy plugins properly instead of leaking the ones known to crash on teardown (debugging)
LC_ALL / LC_MESSAGES / LANG	The interface language at first start, until one is chosen in Settings
LV2_PATH, VST_PATH, VST3_PATH, CLAP_PATH, LADSPA_PATH, DSSI_PATH, SF2_PATH, SFZ_PATH	Replace the search path for that format
File in ~/.local/state/choz/	What it holds
choz.log	The log
plugins.json	The scan cache: what was found, where, and when
plugin-paths.json	The search folders per format
plugin-verdicts.json	Which plugin passed quarantine and which did not
plugin-sandbox.json	Which plugins are hosted out of process
ui.json	Interface preferences: theme, wallpaper, language, panel opacity

16. Troubleshooting

Symptom	What it usually is
No sound, and MONITOR shows no MIDI	The port is not bound. Open IN (F2), select the port, press Enter — the row should say → tab n . If the port is not listed at all, press r to rescan.
MONITOR shows MIDI but nothing sounds	The tab has no instrument (press 1), the tab is muted or soloed away in the MIXER, or in MULTI the tab is listening on another channel (F4).
A DAW cannot see choz	For MIDI, look for choz MIDI IN in the DAW's MIDI <i>output</i> list — it is a destination, not a source. If it is missing, it was switched off in the IN drawer. For audio, the backend must be JACK (Settings → AUDIO) and choz must be restarted after changing it.
Clicks and dropouts	Raise the buffer size in Settings → AUDIO, or lower the DSP load by bypassing effects. The DSP figure on the menu bar says which of the two you are short of.
Dropouts, but DSP is low	The graph itself is dropping cycles, and the cause is outside choz: another client, or another card xrunning in the same graph. choz says so in the log when it detects it; <code>journalctl --user -q grep spa.alsa</code> names the device.
Everything is slow when built from source	A debug build renders roughly ten times dearer than a release one. Build with <code>--release</code> .
A DAW still shows an old choz	A copy of <code>choz.clap</code> or <code>choz-rack.clap</code> in <code>~/</code> .clap wins over the packaged one in <code>/usr/lib/clap</code> . Remove or rename it (the packages set it aside as <code>*.bak</code>), quit the DAW completely and rescan (section 2.5).
The arranger's band clips	A new tab is trimmed on a middle C of its first preset, and a quiet piano can leave it at ×2 — which a whole band then pushes past full scale. Pull the tab's VOL down, or press n in the MIXER to trim it again while the band plays.
 on the arranger takes a moment	It waits for the metronome's next downbeat — at most a bar. Press on the one to start that bar at once.
The click and the band count different bars	FOLLOW ARR is off (the metronome's menu), or the bar was set by hand, which turns it off. Switch it back on and the click takes the band's meter while it plays.
choz disappeared when a plugin crashed	Run that plugin isolated: x for an instrument, X for an effect, or <code>CHOZ_SANDBOX_GUI=1</code> for every plugin with a window. A plugin that has crashed once is remembered and hosted out of process from then on.
A plugin is not in the picker	Its folder is not on the search path: Settings → Plugin Paths, then add it and rescan (EDIT → Rescan plugin paths, or F5 in the picker). A format's environment variable, when set, replaces the whole list for that format.
A plugin window does not open	LADSPA, DSSI and Pure Data have no embeddable window by design — use the knobs in the rack. Otherwise choz needs X11 or XWayland.
A microphone tab howls	The feedback guard cuts the loop; leave it on (Settings → AUDIO). Lower IN on the tab and check that the tab is not routed into what the microphone hears.
The wallpaper is missing or flickers	Outside kitty, Ghostty and WezTerm it is painted with cell colours. <code>CHOZ_KITTY_BG=0</code> forces that path everywhere; Clear background turns it off.
The tempo drifts or will not change	Something else is driving the clock: check CLK on the menu bar. On an external source, the tempo is the sender's and TAP will not hold.

17. Glossary

Term	Meaning in this guide
Rack	The middle of the screen: all the tabs, and what they hold.
Tab	One channel of the rack: an input, an instrument, an effect chain, an arpeggiator, a sequencer, a fader and a destination.
Drawer	The IN and OUT lists at the screen's edges (F2, F3).
Direct out	A pair of graph ports belonging to one tab, off the master mix, for a DAW to record.
Group	One of four submixes (A–D) between the tabs and the main mix.
Sandbox	Running a plugin in a child process so its crash cannot take choz down.
Quarantine	The probe choz runs on an unknown plugin before hosting it in its own process.
Gate (on an effect)	Another tab driving this effect's dry/wet or a knob, with its level or its notes.
A → M	Audio to MIDI: turning what an audio tab hears into notes.
Quantum	PipeWire's block size, which choz asks for from the configured buffer.
xrun	A cycle the audio graph failed to deliver on time; heard as a click.

Notes on this edition. The screenshots come from choz 1.3.9 running with the plugins installed on that machine; the screen is unchanged in 1.3.10 apart from the TAP button described in section 10.3, which sits immediately left of MET. Plugin names, banks and presets shown are examples, not requirements.